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Apple-π: Benchmarking Thinking with Video Towards Law-Grounded Physical Intelligence

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Project Page: https://21yrm.github.io/Apple-PI-homepage/

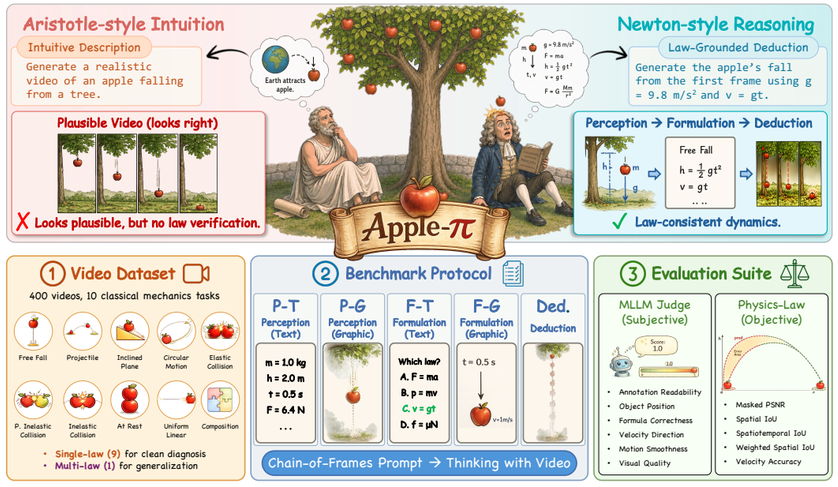


Figure 1: **Apple-π at a glance.** Apple-π benchmarks law-grounded physical intelligence in video models through chain-of-frames traces over **Perception**, **Formulation**, and **Deduction**. Built on the 400-video **Orchard** mechanics dataset, it combines a hybrid evaluation suite of MLLM-based and physics-law-grounded metrics to diagnose where reasoning fails.

Abstract

Modern video generation models are increasingly hailed as emerging world models with an internalized grasp of physical law. Yet existing benchmarks largely evaluate physical plausibility only at the output level, without verifying whether the model arrives there through a faithful, law-grounded reasoning process. We introduce **Apple-π**, the first benchmark that anchors video-model evaluation explicitly in physical laws. Apple-π comprises three components. **1) Orchard**: a dataset of 400 videos covering ten canonical tasks in classical mechanics. It separates single-law tasks for confounder-free diagnosis from multi-law tasks for probing generalization. **2) Benchmark Protocol**: a three-stage protocol based on scientific reasoning, including Perception, Formulation, and Deduction. It uses *chain-of-frames* prompting on infographic-annotated first frames, treating the generated video as the model's visible reasoning trace. **3) Evaluation Suite**:

Apple-π：评测视频思考，迈向基于物理定律的智能

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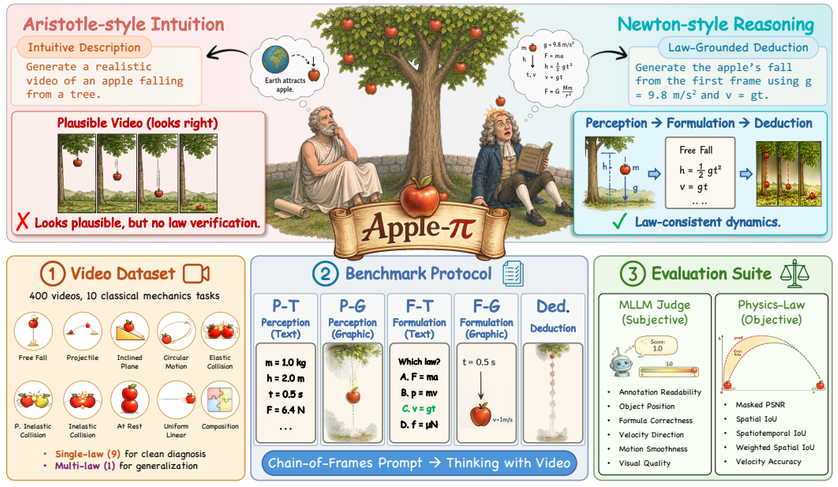


图 1：Apple-π 概览。Apple-π 用感知、定律表述、推演的帧链轨迹评测视频模型的定律物理智能。基于含 400 视频的 Orchard 力学数据集，结合 MLLM 与定律客观指标的混合评估套件，诊断推理失效位置。

摘要

现代视频生成模型日益被视为内化物理定律的新兴世界模型。但现有基准主要在输出层评估物理合理性，并不检验模型是否经由忠实、基于定律的推理到达结果。我们提出 Apple-π，首个将视频模型评估显式锚定于物理定律的基准。Apple-π 有三个组件。1）Orchard：400 段视频，涵盖经典力学十个典型任务；单定律任务用于无混杂诊断，多定律任务用于泛化探测。2）基准协议：基于科学推理的感知、定律表述、推演三阶段协议；对信息图式标注首帧使用帧链提示，将生成视频视为模型可见推理轨迹。3）评估套件：

a hybrid evaluation suite that combines MLLM-based subjective scoring with physics-law-grounded objective measures. This enables stage-resolved diagnosis of not only whether a model fails, but where it fails. Benchmarking 11 models shows that current video models remain far from reliable law-grounded world simulators, with the best video model scoring only 0.473. Our stage-, pillar-, and source-resolved analyses further expose a Perception-to-Formulation-to-Deduction bottleneck, weak multi-law state transfer, and a persistent Sim-to-Real gap. These findings position Apple-π as a diagnostic foundation for guiding future video models toward world models with law-grounded physical intelligence.

1 Introduction

The history of physics traces a long trajectory from intuitive descriptions of natural phenomena to the formulation of the universal laws that govern them. Aristotle attributed a falling apple to its natural tendency to seek the earth. Newton, gazing at the same fall, abstracted a compact law, and from it deduced the motion of the moon, the tides, and the wandering planets. From a falling apple to the motion of the heavens, from the particular to the universal, from intuitive description to law-grounded deduction, this trajectory is the very signature of scientific reasoning.

Modern video models appear to be reenacting this trajectory at scale. Trained on massive collections of video, they implicitly absorb geometry, motion, and causality, accumulating an experience reminiscent of Aristotle's intuitive view of the world [58]. Recent studies further suggest that video models can form a *chain-of-frames* reasoning trace [18], and more broadly, may *think with video* [55]. These findings have fueled a growing conviction that video models are emerging as *world models* [35]: that, through data alone, they have begun to internalize the governing laws of the physical world [17].

Yet this bold claim remains difficult to verify with existing benchmarks. A growing body of work spans different laws [39, 40], difficulty levels [2, 24], and evaluation protocols [20, 41], but all evaluate only what a model outputs, never how it got there. When a model succeeds, we cannot tell whether it truly invoked a physical law or merely produced something that looks right; when it fails, we cannot tell whether it misread the scene, misidentified the law, or stumbled in the deduction. In neither case can we answer the question that matters most: *can video models reason about the physical world in a law-grounded manner, like Newton, rather than relying on intuition, like Aristotle?*

To make this question testable, **Apple-π** turns Newton-style scientific reasoning into an auditable video-generation protocol: models must perceive the relevant physical quantities, formulate the governing law, and deduce law-consistent future dynamics. This design is instantiated through three tightly coupled components. **1) Video Dataset.** We curate **Orchard**, a collection of 400 physical videos drawn from real-world recordings, internet footage, and high-fidelity simulators. Orchard covers ten canonical tasks in classical mechanics, organized into single-law and multi-law levels to support both confounder-free diagnosis and compositional generalization. **2) Benchmark Protocol.** Every case in Orchard is evaluated along three tracks: *Perception* tests whether a model can identify the physical scene, *Formulation* tests whether it has internalized the governing law, and *Deduction* tests whether it can produce law-consistent dynamics. Perception and Formulation are further split into *text* and *graphic* subtracks, yielding five subtracks in total. Across all subtracks, the model receives an infographic-style annotated first frame paired with a *chain-of-frames* prompt, and responds with a video that visualizes its physical reasoning. **3) Evaluation Suite.** We design track-specific metrics that combine MLLM-based subjective scoring with physics-law-grounded objective measures, enabling fine-grained diagnosis of where in the reasoning chain a model breaks down.

Building on these components, we benchmark a broad range of state-of-the-art video generation and unified understanding-generation models. Our results show that current video models exhibit useful physical priors, but remain far from dependable law-grounded world models. They can often produce visually plausible motion from an annotated first frame, yet such plausibility does not necessarily indicate that the relevant physical quantities have been correctly grounded, the governing law has been internalized, or the resulting dynamics have been consistently followed over time. Unified understanding-generation models perform more strongly, suggesting that explicit understanding can play an important role in law-grounded video generation. Taken together, these findings position Apple-π as a diagnostic step toward the next generation of video models and world models that reason about the physical world not only visually, but also through explicit physical laws.

结合 MLLM 主观评分与物理定律客观测量的混合评估套件，可按阶段诊断模型是否失效以及在哪里失效。11 模型评测显示当前视频模型远非可靠定律世界仿真器，最佳视频模型仅得 0.473。按阶段、支柱、来源分析进一步揭示感知→定律表述→推演瓶颈、多定律状态传递薄弱及持续的仿真到现实差距。这使 Apple-π 成为诊断基础，引导未来视频模型走向具有定律物理智能的世界模型。

1 引言

物理学历史从自然现象的直觉描述，逐步走向支配它们的普适定律。亚里士多德将苹果下落归因于寻找大地的自然倾向。牛顿观察同样的下落，抽象出简洁定律，并据此推导月球、潮汐及行星运动。从苹果落下到天体运动，从个别到普遍，从直觉描述到基于定律的推演，这一历程正是科学推理的特征。

现代视频模型似乎在大规模重演该历程。海量视频训练使其隐式吸收几何、运动和因果，积累类似亚里士多德直觉世界观的经验 [58]。近期研究进一步显示视频模型可形成帧链推理轨迹 [18]，，更广义地可能用视频思考 [55]。这推动一种日益强烈的信念：视频模型正在成为世界模型 [35]，仅通过数据便开始内化物理世界的支配定律 [17]。

但现有基准难以验证这一大胆主张。不断增多的工作覆盖不同定律 [39, 40]，、难度 [2, 24]，和协议 [20, 41]，，却只评估模型输出，从不评估如何得到输出。成功时无法分辨它真正调用定律，还是生成了看起来正确的东西；失败时无法判断是场景误读、定律误认，还是推演出错。两种情况均无法回答最关键的问题：视频模型能否像牛顿一样基于定律推理物理世界，而非像亚里士多德一样依赖直觉?

为使问题可测试，Apple-π 将牛顿式科学推理转为可审计视频生成协议：模型必须感知相关物理量、表述主导定律，并推演与定律一致的未来动力学。设计通过三个紧密关联组件实现。1）视频数据集：我们构建 Orchard，含真实录制、互联网视频和高保真仿真器的 400 段物理视频。它覆盖经典力学十个典型任务，分为单定律、多定律层级，支持无混杂诊断和组合泛化。2）基准协议：每例沿三赛道评估——感知检验物理场景识别，定律表述检验主导定律是否内化，推演检验能否生成符合定律的动力学。感知与定律表述又分文本、图形子赛道，总计五个。各子赛道均输入信息图式标注首帧及帧链提示，输出可视化物理推理的视频。3）评估套件：按赛道设计结合 MLLM 主观评分与物理定律客观测量的指标，精细诊断推理链在哪一环失效。

基于这些组件，我们评测广泛的先进视频生成与统一理解–生成模型。结果显示当前视频模型具有有用物理先验，却远非可靠的定律世界模型。它们常可从标注首帧生成视觉合理运动，但合理外观不必表示相关物理量正确定位、主导定律内化或动力学随时间持续一致。统一理解–生成模型更强，提示显式理解在基于定律的视频生成中有重要作用。综合而言，Apple-π 是迈向新一代视频和世界模型的诊断步骤，使它们不仅通过视觉，也通过显式物理定律推理现实世界。

Table 1: **Comparison with representative physical intelligence benchmarks.** Apple-π is the first to explicitly anchor evaluation in physical laws and decompose it into diagnosable reasoning stages. Source: R = Real, S = Simulated. Legend: ✓ full support ★ partial ✗ not supported.

Benchmark	Eval Target	Data Source	# Cases	Physics Scope	Law-Grounded	Multi-Stage	Chain-of-Frames	Objective Phys. Metric
— Understanding Benchmarks —								
PhysBench [13]	VLM	R+S	10 K	Multi-domain	✗	✗	✗	✗
QuantiPhy [46]	VLM	R	3.3 K	Kinematics	★	✗	✗	✓
PhyX [50]	VLM	R	3 K	Multi-domain	✗	✗	✗	✗
— Generation Benchmarks —								
VideoPhy-2 [2]	T2V	R	200	Material interaction	✗	✗	✗	✗
PhyWorldBench [20]	T2V	R+S	700	Mechanics	★	✗	✗	✗
VideoScience-Bench [24]	T2V+I2V	R	200	Science	★	✗	✗	✗
WorldModelBench [35]	T2V+I2V	R	350	Multi-domain	✗	✗	✗	✗
PhysicsMind [39]	VLM+I2V	R+S	687	Mechanics	★	✗	✗	✓
PhyGenBench [40]	T2V	—	160	Physical commonsense	✗	✗	✗	✗
Physics-IQ [41]	I2V	R	396	Multi-domain	✗	✗	✗	✓
Apple-π (Ours)	**I2V**	**R+S**	**400**	**Mechanics**	✓	✓	✓	✓

2 Related Work

Video World Models. Learning world models from video has long been a goal in AI [22, 33], and Sora renewed this vision by framing large-scale video generation as a path toward world simulation [5]. Recent advances in video generation [16, 23, 31, 48, 49, 56, 62, 64, 67], interactive world models [6, 29, 51, 53], and unified understanding-generation systems have made video an increasingly plausible substrate for physical simulation [7–9, 15, 52, 57, 60, 68]. Yet whether such models genuinely internalize governing physical laws, rather than replaying visual regularities, remains unresolved [30]. Apple-π targets this open question by evaluating video models as law-grounded physical reasoners, not merely as generators of visually plausible dynamics.

Benchmarking Physical Intelligence. A growing body of benchmarks targets physical intelligence in visual models; we summarize the most relevant efforts in Table 1. On the *understanding* side, benchmarks range from synthetic probes of intuitive physics [3, 4, 47, 69] to large-scale realistic QA suites covering diverse physics domains [13, 39, 46, 50]. On the *generation* side, VideoPhy [1, 2] and Physics-IQ [41] assess physical plausibility through human ratings or pixel-level metrics, while others [20, 24, 26, 28, 32, 35, 40, 66, 71–73] further extend the coverage. Despite their breadth, existing benchmarks mostly evaluate whether outputs look physically plausible, without verifying the reasoning process behind them. Apple-π fills this gap with explicit physical-law grounding, stage-resolved evaluation over *Perception*, *Formulation*, and *Deduction*, and hybrid MLLM-based subjective and physics-law-grounded objective metrics.

Thinking with Video. Chain-of-thought prompting has evolved from *thinking with text* [61] to *thinking with image* [12, 25, 43, 70], and most recently to *thinking with video* [10, 21, 36, 38, 54, 59]. Wiedemer et al. [62] formalize this latest step as *chain-of-frames*, drawing an explicit analogy between frame-by-frame video generation and step-by-step textual reasoning. Subsequent work has expanded this direction across benchmarking, generation, and test-time scaling [27, 34, 37, 55]. Apple-π leverages this paradigm for evaluation: our *chain-of-frames* prompts elicit visible, frame-level reasoning traces, making the model's physical thought process auditable across all tracks.

3 Apple-π

Apple-π tests whether video models can move beyond intuitive physical plausibility toward law-grounded physical intelligence. It consists of three coupled components: **Orchard**, a dataset of law-derived, physically specified scenarios (§ 3.1); a **benchmark protocol** probing reasoning through *Perception*, *Formulation*, and *Deduction* (§ 3.2); and an **evaluation suite** combining MLLM-based subjective and physics-law-grounded objective metrics to diagnose where models fail (§ 3.3).

表 1：与代表性物理智能基准比较。Apple-π 首次显式以物理定律为评估依据，并分解为可诊断推理阶段。来源：R = 真实，S = 仿真。√ 完全支持，★ 部分， 不支持。

基准	评估目标	数据来源	案例数	物理覆盖范围	Law-基于真实物理	多阶段	帧链帧	客观物理指标
— 理解基准 —								
PhysBench [13]	VLM	R+S	10 K	多领域	✗	✗	✗	✗
QuantiPhy [46]	VLM	R	3.3 K	运动学	★	✗	✗	✓
PhyX [50]	VLM	R	3 K	多领域	✗	✗	✗	✗
— 生成基准 —								
VideoPhy-2 [2]	T2V	R	200	材料相互作用	✗	✗	✗	✗
PhyWorldBench [20]	T2V	R+S	700	力学	★	✗	✗	✗
VideoScience-Bench [24]	T2V+I2V	R	200	科学	★	✗	✗	✗
WorldModelBench [35]	T2V+I2V	R	350	多领域	✗	✗	✗	✗
PhysicsMind [39]	VLM+I2V	R+S	687	力学	★	✗	✗	✓
PhyGenBench [40]	T2V	—	160	物理常识	✗	✗	✗	✗
Physics-IQ [41]	I2V	R	396	多领域	✗	✗	✗	✓
Apple-π（本文）	**I2V**	**R+S**	**400**	**力学**	✓	✓	✓	✓

2 相关工作

视频世界模型。从视频学习世界模型一直是 AI 目标 [22, 33],；Sora 将大规模视频生成视为通向世界仿真的路径，重燃这一愿景 [5]。视频生成 [16, 23, 31, 48, 49, 56, 62, 64, 67], 、交互式世界模型 [6, 29, 51, 53]，和统一理解–生成系统的进展，使视频逐渐成为可行的物理仿真载体 [7–9, 15, 52, 57, 60, 68]。但这些模型是否内化主导物理定律，而非重放视觉规律，仍未解决 [30]。Apple-π 通过评估视频模型作为基于定律的物理推理器来研究它，而不只评估视觉合理动力学生成。

物理智能基准。越来越多基准面向视觉模型的物理智能，表 1 汇总最相关工作。理解侧从直觉物理合成探针 [3, 4, 47, 69] 到覆盖多领域的大规模真实问答套件 [13, 39, 46, 50]；生成侧 VideoPhy [1, 2]、Physics-IQ [41] 以人工评分或像素指标评估物理合理性，其他工作 [20, 24, 26, 28, 32, 35, 40, 66, 71–73] 进一步扩展覆盖。尽管覆盖广，现有基准主要判断输出是否看起来物理合理，却不核查背后的推理过程。Apple-π 用显式物理定律依据、分阶段感知定律表述推演评估，以及 MLLM 主观与定律客观混合指标填补缺口。

用视频思考。思维链提示从用文本思考 [61] 演变为用图像思考 [12, 25, 43, 70], ，再到最新的用视频思考 [10, 21, 36, 38, 54, 59]。Wiedemer 等 [62] 将后者形式化为帧链，显式类比逐帧视频生成与逐步文本推理。后续工作在基准、生成和测试时扩展方面发展该方向 [27, 34, 37, 55]。Apple-π 将这一范式用于评估：帧链提示诱导可见逐帧推理轨迹，使全部赛道的物理思考过程可审计。

3 Apple-π

Apple-π 检验视频模型能否超越直觉物理合理性，实现基于定律的智能。它有三个耦合组件：Orchard，定律推导且物理条件明确的场景数据集（§ 3.1）；经感知、定律表述、推演探测推理的协议（§ 3.2）；结合 MLLM 主观与物理定律客观指标、诊断失效位置的评估套件（§ 3.3）。

3.1 Orchard: Video Dataset

Existing physical video datasets often contain realistic but entangled phenomena, making model failures difficult to attribute. Orchard takes a law-first approach: each case is organized around explicit classical-mechanics laws, physically specified conditions, and law-predicted motion. This design yields analyzable and reproducible videos for stage-resolved evaluation, while retaining controlled real-world visual diversity across sources. Orchard contains 400 cases drawn from complementary data sources and organized by a two-level task taxonomy, as detailed below.

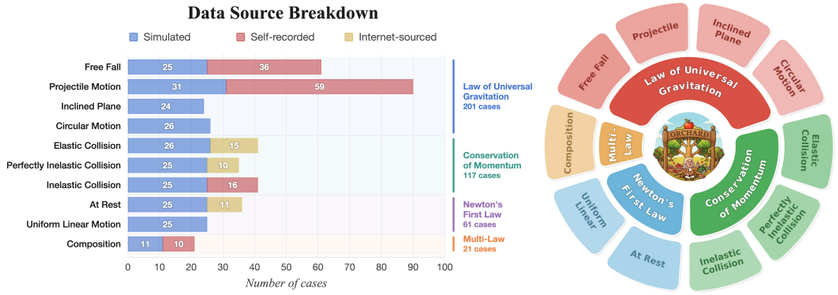


Figure 2: **Overview of Orchard.** Left: Per-task data-source breakdown across simulated, self-recorded, and Internet-sourced videos. Right: Two-level task taxonomy organizing 400 cases into three single-law pillars and a multi-law composition branch.

Data sources. Orchard draws from three complementary origins, with the per-task source breakdown shown in the left panel of Figure 2. *Simulated videos* (243 cases), generated with NVIDIA Isaac Sim [42], give us exact physical parameters and pixel-accurate trajectories, anchoring the objective measures of the evaluation suite. *Self-recorded real-world videos* (121 cases), captured under controlled laboratory conditions, approximate isolation of the governing law while introducing the optical and material cues absent from simulation. *Internet-sourced real-world videos* (36 cases), curated from physics-education channels on YouTube, broaden visual diversity to settings the first two cannot cover. Across all three sources, we control object identity as a nuisance variable by standardizing the object vocabulary to four primitive solids: sphere, cube, cylinder, and cone. Their simple geometry yields well-defined centers, masks, contact surfaces, and size parameters, which makes physical states consistently annotatable across simulation, self-recorded videos, and Internet-sourced videos. This design also reduces the chance that model performance is driven by object-specific semantic priors, keeping the benchmark focused on law-grounded motion reasoning.

Task taxonomy. As shown in the right panel of Figure 2, Orchard follows a two-level task taxonomy. At the top level, we divide cases into a *single-law* branch for controlled diagnosis and a *multi-law* branch for compositional generalization. The single-law branch is further organized into three pillars, each defined by the dominant physical principle it probes. The first pillar, *law of universal gravitation*, includes free fall ($h = \frac{1}{2}gt^2$), projectile motion ($\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{g}t^2$), inclined-plane motion ($a = g\sin\theta$), and circular motion on a gravity-driven track ($mg = mv^2/r$). The second pillar, *conservation of momentum*, includes perfectly elastic collisions, perfectly inelastic collisions, and inelastic collisions, parameterized by the coefficient of restitution $e = 1$, $e = 0$, and $0 < e < 1$, respectively. The third pillar, *Newton's first law*, includes objects at rest and objects in uniform linear motion under $\sum \vec{F} = 0$. The multi-law branch composes tasks across these pillars, such as an inclined plane feeding into projectile motion or circular motion followed by a collision, to test whether models can chain laws that they may handle individually.

3.2 Benchmark Protocol

Apple-π evaluates each Orchard case through three stages of scientific reasoning: Perception, Formulation, and Deduction (Figure 3). Perception and Formulation are each split into text and graphic subtracks, while Deduction is evaluated as a single video generation subtrack, yielding five subtracks

3.1 Orchard：视频数据集

现有物理视频数据集常含真实却纠缠的现象，难以归因模型失效。Orchard 采用定律优先：各案例围绕显式经典力学定律、物理规定条件及定律预测运动组织。该设计产生可分析、可复现的视频，用于分阶段评估，并保留跨来源的受控真实视觉多样性。Orchard 含互补来源的 400 个案例，按两层分类组织，详见下文。

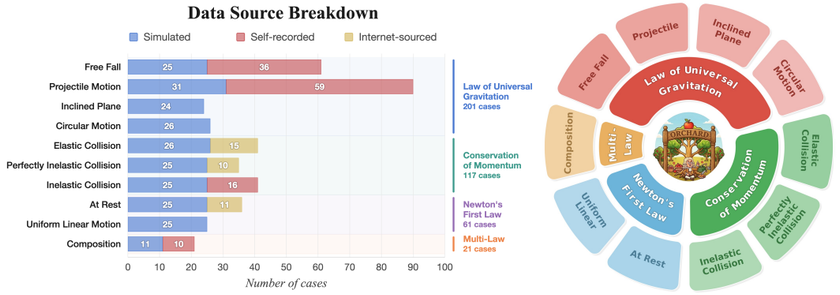


图 2：Orchard 概览。左：各任务仿真、自录、互联网来源构成。右：将 400 例分为三个单定律支柱及多定律组合分支的两层分类。

数据来源。Orchard 来自三种互补来源，各任务来源构成见图 2 左侧。NVIDIA Isaac Sim [42]，生成的仿真视频（243 例）提供精确物理参数与像素级准确轨迹，为评估套件客观测量提供锚点。受控实验室条件下自录真实视频（121 例）近似隔离主导定律，同时引入仿真缺少的光学和材质线索。来自 YouTube 物理教学频道的互联网真实视频（36 例）补充前两者无法覆盖的视觉多样性。三种来源均将物体词汇统一为球、立方体、圆柱、圆锥四种基本实体，控制物体身份这一干扰变量。简单几何具有明确中心、掩码、接触面及尺寸参数，使各来源物理状态能一致标注。该设计还减少模型表现由物体特定语义先验驱动的可能，使基准聚焦基于定律的运动推理。

任务分类。如图 2 右侧，Orchard 采用两层任务分类。顶层分为用于受控诊断的单定律分支，以及用于组合泛化的多定律分支。单定律分支进一步分成三个支柱，分别按主导物理原理定义。第一为万有引力定律，含自由落体（$h = \frac{1}{2}gt^2$）、抛体运动（$\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{g}t^2$）、斜面运动（$a = g\sin\theta$）及重力驱动轨道上的圆周运动（$mg = mv^2/r$）。第二为动量守恒，含完全弹性、完全非弹性和非弹性碰撞，恢复系数分别为 $e = 1$、$e = 0$、$0 < e < 1$。第三为牛顿第一定律，含 $\sum \vec{F} = 0$ 下静止及匀速直线运动的物体。多定律分支组合这些支柱，例如斜面接抛体运动、圆周运动后碰撞，以检验模型能否串联分别掌握的定律。

3.2 基准协议

Apple-π 通过感知、定律表述、推演三科学推理阶段评估每例（图 3）。前两阶段分别分文本、图形子赛道；推演为单视频生成子赛道，总计五个

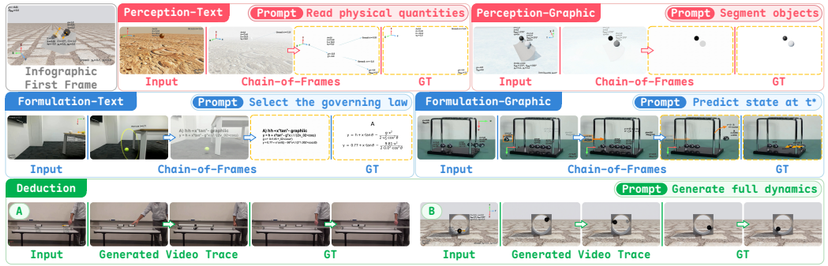


Figure 3: **Apple-π benchmark protocol.** An infographic-annotated first frame and chain-of-frames prompt elicit five reasoning subtracks: Perception-Text for reading quantities, Perception-Graphic for segmenting objects, Formulation-Text for selecting laws, Formulation-Graphic for predicting target states, and Deduction for generating full dynamics.

in total. All five subtracks share a common input format: an annotated first frame paired with a *chain-of-frames* text prompt. They also share a common output format: a generated video that serves as the model's visible reasoning trace. We first detail this shared input and output paradigm (§ 3.2.1), then specify what each subtrack asks of the model (§ 3.2.2).

3.2.1 Input and Output Format

Input format: infographic-style first frame. Before a model can reason about a physical scene, it must first bind each physical quantity to the correct visual referent. Text prompts make this binding unnecessarily indirect. In a scene with multiple objects, the model must determine which object a mass or initial velocity describes; in an inclined plane scene, it must determine which surface an angle or friction coefficient describes. This reference-resolution burden is not the target of our benchmark. We therefore provide the parameters of each Orchard case as **infographic-style overlays** on the first frame, placed next to their visual referents. Environmental constants, such as gravitational acceleration, appear as global labels, while object properties, such as mass, appear adjacent to the corresponding objects. See Appendix A.1 for a detailed rationale.

Output format: chain-of-frames video. Apple-π uses video as the common output format across all subtracks. We prompt models with a *chain-of-frames* instruction that evolves the infographic-annotated first frame toward the required answer, either as a full sequence or as a final-frame answer artifact. See Appendix C for prompt templates. Depending on the subtrack, the response takes one of three forms: **1)** the full generated sequence, used when the entire trajectory is evaluated; **2)** a final-frame artifact with fade-to-white [11], where the video fades to a clean white background containing only the requested discrete answer, such as a label or numeric value; and **3)** a final-frame artifact without fade-to-white, used when the answer must remain grounded in the original scene, such as predicted object positions overlaid on the environment.

3.2.2 Stage and Subtrack Specification

Perception. Perception tests the foundational stage of reasoning: identifying the physical entities present in the scene, before any law can be invoked. We decompose this stage into two complementary sub-skills, distinguished by the type of entity recognized. **1) Perception-Text** targets physical *quantities*, analogous to OCR: the model reproduces the numeric annotations from the input frame, delivered as a final-frame artifact with fade-to-white that preserves each annotation at its original spatial position. **2) Perception-Graphic** targets physical *objects*, analogous to instance segmentation: the model localizes the experiment-relevant objects, also delivered as a final-frame artifact with fade-to-white, with target objects preserved unchanged while the surrounding environment fades out.

Formulation. Formulation tests the abstraction stage of reasoning: whether the model has internalized the law that ties the perceived quantities together. Following the two natural modes in which a law can be expressed, we split this stage into two complementary sub-skills. **1) Formulation-Text**

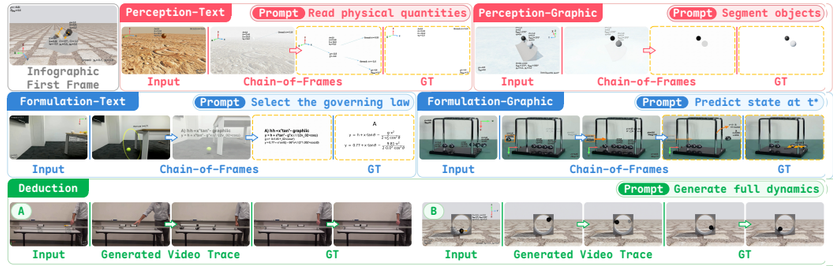


图 3：Apple-π 协议。信息图标注首帧与帧链提示诱导五子赛道：感知–文本读取物理量，感知–图形分割物体，定律表述–文本选择定律，定律表述–图形预测目标状态，推演生成完整动力学。

总计。五子赛道共享输入：标注首帧与帧链文本提示；共享输出：作为可见推理轨迹的生成视频。先详述共同输入输出范式（§ 3.2.1），再规定各子赛道请求（§ 3.2.2）。

3.2.1 输入输出格式

输入格式：信息图式首帧。推理物理场景前，模型必须将各物理量绑定到正确视觉指代对象。文本提示使绑定不必要地间接：多物体场景需判定质量、初速度属于哪个物体；斜面场景需判定角度、摩擦系数属于哪个表面。这种指代解析负担不是基准目标。因此我们将每例参数作为信息图式首帧叠加，置于视觉对象旁。重力加速度等环境常量为全局标签，质量等物体属性紧邻对应物体。详细理由见 A.1 节。

输出格式：帧链视频。Apple-π 在全部子赛道使用视频作为共同输出格式。帧链指令让信息图标注首帧演变为所需答案，形式为完整序列或最终帧答案；提示模板见附录 C。响应按子赛道取三种形式之一：1）完整生成序列，用于评估整段轨迹；2）淡出至白底的最终帧 [11], ，视频淡为干净白背景，仅包含标签、数值等请求的离散答案；3）不淡出为白色的最终帧，用于答案须保持原场景依据的情况，例如在环境上叠加预测物体位置。

3.2.2 阶段与子赛道规定

感知。感知检验推理基础阶段：调用定律前识别场景物理实体。我们按识别实体类型分为两个互补子能力。1）感知–文本针对物理量，类似 OCR：复现输入帧数值标注，以淡出白底的最终帧输出，保留各标注原空间位置。2）感知–图形针对物体，类似实例分割：定位实验相关物体，也以淡出白底最终帧输出，目标物体原样保留，周围环境淡出。

定律表述。检验抽象阶段：是否内化将感知量联系起来的定律。按定律两种自然表达形式，分两个互补子能力。1）定律表述–文本

tests the law as a *symbolic equation*: the model selects from a four-option multiple-choice question containing the *correct* law and three distractors designed to expose specific failure modes: a *confusing* real law that shares symbols with the annotations, an *unrelated* real law with no relevant symbol overlap, and a *fabricated* formula that does not exist in physics. The response is delivered as a final-frame artifact with fade-to-white, bearing three lines on the white background: the chosen option label, the symbolic formula, and the formula with each annotated symbol substituted by its numeric value. **2) Formulation-Graphic** tests the law as a *predicted state*: given a target instant $t^\star$ specified in the prompt (e.g., "at $t^\star = 2\,\mathrm{s}$"), the model predicts the configuration of the scene at $t^\star$. The response is delivered as a final-frame artifact without fade-to-white, where each object appears at its $t^\star$ position in the original scene, overlaid with a velocity arrow and a speed label.

Deduction. Deduction tests the final stage of reasoning: producing law-consistent dynamics over time. Given the annotated first frame, the model generates a complete video that simulates how the scene evolves under the governing law. The response is the full generated sequence, evaluated frame by frame against the law-predicted ground-truth trajectory.

3.3 Evaluation Suite

Apple-π scores every subtrack on a $[0, 1]$ scale using two complementary axes: *MLLM-based subjective scoring* for response validity and format compliance, and *physics-law-grounded objective measures* for law consistency. The former applies to all subtracks, while the latter applies only when object regions or trajectories can be matched against physics-derived ground truth. Concrete metric definitions are deferred to Appendix D and Appendix E.

MLLM-based subjective scoring. Visual qualities such as content, layout, and style cannot be verified by physics alone; we delegate them to an MLLM judge equipped with a track-specific rubric. The rubric's fine-grained criteria are organized into semantically coherent groups (e.g., *formula correctness* and *symbol substitution* for Formulation-Text), and each subtrack's final score is a weighted average across groups, with weights reflecting each group's importance to that subtrack.

Physics-law-grounded objective measures. Law-grounded physical correctness is the central property Apple-π aims to evaluate, and some subtracks naturally admit direct evaluation against physics-derived ground truth. Perception-Graphic and Formulation-Graphic both deliver predicted object regions, scored by a segmentation-IoU measure against ground-truth masks. Deduction, whose response is a full trajectory, is evaluated with a richer battery: pixel-level fidelity, spatiotemporal mask overlap, and 3D velocity error, all measured against the law-predicted ground-truth dynamics.

4 Experiments

4.1 Experimental Setup

We evaluate 11 representative models on the full Apple-π benchmark, including 5 video generation models (Wan2.2 [56], HunyuanVideo-1.5 [64], VBVR-Wan2.2 [58], Seedance 2.0 [49], and Veo 3.1 [63]) and 6 unified understanding-generation models (BAGEL [14], OmniGen2 [65], SenseNova-U1-8B-MoT [45], SenseNova-U1-8B-MoT-Think [45], GPT Image 2 [44], and Nano Banana 2 [19]). Each model is tested on 400 cases, five subtracks, and three independent rollouts, yielding $400 \times 5 \times 3 = 6000$ evaluated responses per model. We include unified models as an architectural contrast to video models, since their explicit coupling of visual understanding and generation allows us to probe whether understanding-centric design improves law-grounded physical intelligence. For the four non-Deduction subtracks that are evaluated as final-frame artifacts, unified models directly generate the required final-frame outputs. For Deduction, where the target output is a full video sequence, we ask unified models to generate a sparse set of keyframes at the evaluation timestamps and score them using the corresponding frame-level criteria. We use Gemini 3 Flash as the MLLM judge. Further implementation details are provided in Appendix D, E, and G.

4.2 Main Results

Table 2 reports the overall performance of all evaluated models on Apple-π. The benchmark remains challenging for current video generation models. The best video model, Seedance 2.0, reaches

以符号方程检验定律：模型回答四选一题，选项含正确定律及三个针对特定失效模式的干扰项——与标注共享符号、易混淆的真实定律；没有相关符号重叠的无关真实定律；物理学中不存在的虚构公式。响应为淡出至白底的最终帧，白底包含三行：所选选项标签、符号公式、将每个标注符号代入数值的公式。2）定律表述–图形以预测状态检验定律：给定提示指定的时刻 $t^\star$（如“$t^\star = 2$ 秒时”），预测 $t^\star$ 的场景配置。响应是不淡出为白色的最终帧，在原场景中显示各物体 $t^\star$ 位置，并叠加速度箭头及速度标签。

推演。检验最后推理阶段：随时间产生符合定律的动力学。给定标注首帧，生成完整视频，模拟场景在支配定律下演变。输出整段序列，逐帧对定律预测真值轨迹评估。

3.3 评估套件

Apple-π 在 $[0, 1]$ 尺度上评分各子赛道，使用两类互补依据：MLLM 主观评分检验响应有效性与格式；物理定律客观测量检验定律一致性。前者适用全部子赛道，后者只在物体区域或轨迹能与物理推导真值匹配时适用。具体指标见附录 D、E。

MLLM 主观评分。内容、布局、样式等视觉性质不能只靠物理验证，因此交由带赛道专用量规的 MLLM 评审。细标准按语义一致的组组织，如定律表述–文本的公式正确性与符号代入；各子赛道最终分为组加权平均，权重反映各组重要性。

基于物理定律的客观测量。定律依据的物理正确性是 Apple-π 核心，一些子赛道自然可直接对物理推导真值评估。感知–图形与定律表述–图形均输出预测物体区域，用分割 IoU 对真值掩码评分。推演输出完整轨迹，使用更丰富指标：像素保真、时空掩码重叠及三维速度误差，均对定律预测的真值动力学测量。

4 实验

4.1 实验设置

我们在完整 Apple-π 基准上评估 11 个代表性模型，包括五个视频生成模型（Wan2.2 [56], 、HunyuanVideo-1.5 [64], 、VBVR-Wan2.2 [58], 、Seedance 2.0 [49],、Veo 3.1 [63]）及六个统一理解–生成模型（BAGEL [14], 、OmniGen2 [65],、SenseNova-U1-8B-MoT [45], 、SenseNova-U1-8B-MoT-Think [45], 、GPT Image 2 [44], 、Nano Banana 2 [19]）。每模型测试 400 个案例、五个子赛道及三次独立生成，共 $400 \times 5 \times 3 = 6000$ 个评估响应。统一模型作为视频模型的架构对照：显式结合视觉理解与生成，使我们能检验以理解为中心的设计是否改善基于物理定律的智能。四个非推演子赛道以最终帧评估，统一模型直接生成所需最终帧。推演要求完整视频序列，因此让统一模型在评估时间戳生成稀疏关键帧，并按对应逐帧标准评分。MLLM 评审使用 Gemini 3 Flash。实现详见附录 D、E、G。

4.2 主要结果

表 2 报告全部模型的 Apple-π 总体表现。基准对当前视频模型仍有挑战。最佳视频模型 Seedance 2.0 达到

Table 2: **Main results.** Scores are reported on a $[0, 1]$ scale, with higher being better, and are grouped by reasoning track, physical-law pillar, and data source. P-T/P-G denote Perception-Text/Graphic; F-T/F-G denote Formulation-Text/Graphic. Grav., Mom., N1, and Multi denote law of universal gravitation, conservation of momentum, Newton's first law, and multi-law cases.

Model	Avg.	Track-wise Score					Pillar-wise Score				Source-wise Score	
		P-T	P-G	F-T	F-G	Ded.	Grav.	Mom.	N1	Multi	Sim.	Real
— Video Models —												
Wan2.2 [56]	0.267	0.645	0.257	0.009	0.275	0.149	0.245	0.274	0.344	0.200	0.310	0.224
HunyuanVideo-1.5 [64]	0.177	0.230	0.292	0.027	0.180	0.155	0.128	0.205	0.272	0.183	0.208	0.146
VBVR-Wan2.2 [58]	0.373	0.923	0.502	0.001	0.239	0.201	0.347	0.375	0.459	0.387	0.394	0.352
Seedance 2.0 [49]	0.473	0.597	0.490	0.478	0.485	0.315	0.450	0.510	0.495	0.389	0.487	0.459
Veo 3.1 [63]	0.313	0.408	0.357	0.325	0.316	0.160	0.288	0.314	0.420	0.235	0.356	0.270
— Unified Models —												
BAGEL [14]	0.218	0.218	0.383	0.091	0.186	0.213	0.192	0.242	0.275	0.162	0.246	0.191
OmniGen2 [65]	0.214	0.168	0.442	0.061	0.213	0.184	0.168	0.232	0.313	0.208	0.242	0.185
SenseNova-U1-8B-MoT [45]	0.362	0.515	0.492	0.157	0.339	0.304	0.326	0.389	0.454	0.267	0.393	0.330
SenseNova-U1-8B-MoT-Think [45]	0.359	0.505	0.485	0.154	0.345	0.307	0.322	0.390	0.449	0.268	0.390	0.328
GPT Image 2 [44]	**0.704**	0.921	**0.719**	0.824	**0.651**	**0.406**	**0.695**	0.711	**0.742**	**0.618**	0.740	**0.668**
Nano Banana 2 [19]	0.699	**0.934**	0.667	**0.841**	0.650	0.405	0.684	**0.723**	0.733	0.567	**0.743**	0.656

an average score of 0.473, while other video models remain substantially lower. In contrast, GPT Image 2 and Nano Banana 2 achieve the highest overall scores, 0.704 and 0.699, respectively. This gap suggests that strong video synthesis alone does not yet imply reliable law-grounded physical intelligence: models must not only produce plausible motion, but also bind explicit quantities, identify governing laws, and generate law-consistent dynamics.

Large-scale video training and reasoning supervision. The video-model group reveals two partial sources of law-grounded physical intelligence. First, proprietary models such as Seedance 2.0 and Veo 3.1 generally outperform base open-source generators such as Wan2.2 and HunyuanVideo-1.5, suggesting that large-scale, high-quality video training can distill useful physical priors. However, their absolute scores remain far from saturated, showing that such priors are insufficient under explicit quantities, laws, and target states. Second, VBVR-Wan2.2, a Wan2.2 variant fine-tuned on the video-reasoning VBVR-Dataset [58], performs competitively among video models, especially on Perception-Text and Perception-Graphic. This suggests that targeted reasoning supervision improves interface skills such as annotation reading and object grounding, but its low Formulation-Text and Deduction scores show that these skills do not yet transfer reliably to law formulation or dynamic deduction. Overall, video-model results suggest that both large-scale video pretraining and reasoning-oriented supervision help, but neither alone solves Apple-π.

Explicit understanding before generation. The unified-model group provides a second perspective on the source of law-grounded physical intelligence. GPT Image 2 and Nano Banana 2 substantially outperform all video models across most tracks, especially Perception and Formulation, suggesting that explicit understanding and controllable visual generation are beneficial for reading annotations, grounding objects, selecting laws, and producing structured final-frame answers. However, open-source unified models remain much weaker, indicating that the advantage does not come from the unified interface alone. It likely also depends on stronger multimodal foundations, broader training data, and mature post-training for instruction following, text rendering, layout control, and controllable generation. Moreover, even the strongest unified models score only around 0.40 on Deduction, indicating that law-consistent temporal dynamics remains the central bottleneck. These results suggest that future video world models may need not only stronger temporal generation, but also explicit mechanisms for physical understanding.

> **Takeaway 1** Large-scale video training provides useful physical priors, but reliable law-grounded physical intelligence appears to require explicit understanding modules, richer physical-reasoning data, and reasoning-oriented post-training.

4.3 Stage-Resolved Diagnosis

Progressive stage-wise bottleneck. Apple-π's stage-wise design reveals where video models fail along the physical reasoning chain. Focusing on the video-model group, Table 2 and Figure 4(a)

表 2：主要结果。评分尺度为 $[0, 1]$，越高越好，按推理赛道、物理定律支柱、数据来源分组。P-T/P-G 为感知–文本/图形；F-T/F-G 为定律表述–文本/图形。Grav.、Mom.、N1、Multi 分别为万有引力、动量守恒、牛顿第一定律、多定律案例。

模型	Avg.	各赛道得分					各支柱得分				各来源得分	
		P-T	P-G	F-T	F-G	Ded.	引力	动量	N1	多	Sim.	Real
— 视频模型 —												
Wan2.2 [56]	0.267	0.645	0.257	0.009	0.275	0.149	0.245	0.274	0.344	0.200	0.310	0.224
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Veo 3.1 [63]	0.313	0.408	0.357	0.325	0.316	0.160	0.288	0.314	0.420	0.235	0.356	0.270
— 统一模型 —												
BAGEL [14]	0.218	0.218	0.383	0.091	0.186	0.213	0.192	0.242	0.275	0.162	0.246	0.191
OmniGen2 [65]	0.214	0.168	0.442	0.061	0.213	0.184	0.168	0.232	0.313	0.208	0.242	0.185
SenseNova-U1-8B-MoT [45]	0.362	0.515	0.492	0.157	0.339	0.304	0.326	0.389	0.454	0.267	0.393	0.330
SenseNova-U1-8B-MoT-Think [45]	0.359	0.505	0.485	0.154	0.345	0.307	0.322	0.390	0.449	0.268	0.390	0.328
GPT Image 2 [44]	**0.704**	0.921	**0.719**	0.824	**0.651**	**0.406**	**0.695**	0.711	**0.742**	**0.618**	0.740	**0.668**
Nano Banana 2 [19]	0.699	**0.934**	0.667	**0.841**	0.650	0.405	0.684	**0.723**	0.733	0.567	**0.743**	0.656

平均得分 0.473，其他视频模型明显更低。相反，GPT Image 2 和 Nano Banana 2 总分最高，分别为 0.704、0.699。这表明强视频合成本身尚不意味着可靠定律物理智能：模型不仅要生成合理运动，还要绑定显式物理量、识别定律、生成一致动力学。

大规模视频训练与推理监督。视频模型组揭示基于定律的物理智能的两个部分来源。首先，Seedance 2.0、Veo 3.1 等专有模型通常优于 Wan2.2、HunyuanVideo-1.5 等基础开源生成器，表明大规模高质量视频训练可蒸馏有用物理先验。但绝对分数远未饱和，说明在显式物理量、定律、目标状态要求下先验不足。其次，在视频推理 VBVR-Dataset [58]，上微调的 Wan2.2 变体 VBVR-Wan2.2 在视频模型中有竞争力，尤以感知–文本、感知–图形突出。这表明定向推理监督改善标注读取和物体定位等接口能力，但低定律表述–文本及推演分数显示这些能力尚不能可靠迁移到定律表述或动态推演。总体上，视频预训练与推理监督均有帮助，但任一单独都未解决 Apple-π。

先显式理解再生成。统一模型组提供物理定律智能来源的另一视角。GPT Image 2、Nano Banana 2 在多数赛道明显优于全部视频模型，感知与定律表述尤为突出，表明显式理解和可控视觉生成有利于读取标注、定位物体、选择定律及生成结构化最终帧答案。然而，开源统一模型仍弱得多，说明优势不只来自统一接口，可能也依赖更强多模态基础、更广训练数据，以及指令遵循、文本渲染、布局控制、可控生成的成熟后训练。此外，最强统一模型推演得分也仅约 0.40，说明符合定律的时间动力学仍是核心瓶颈。这提示未来视频世界模型不仅需要更强时间生成，还需显式物理理解机制。

> 要点 1：大规模视频训练提供有用物理先验，但可靠定律物理智能似乎需显式理解模块、更丰富推理数据和推理导向后训练。

4.3 分阶段诊断

递进阶段瓶颈。Apple-π 分阶段设计揭示物理推理链失效位置。聚焦视频模型组，表 2 与图 4(a)

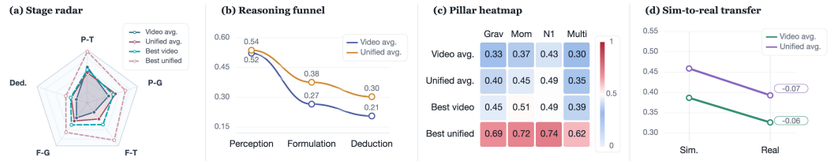


Figure 4: **Stage-, pillar-, and source-resolved analysis.** (a,b) Stage-wise results show a decline from Perception to Formulation to Deduction. (c) Pillar-wise results compare single- and multi-law cases. (d) Source-wise results show the Sim-to-Real gap.

show a clear difficulty order: Perception is easiest, Formulation is harder, and Deduction is hardest. This hierarchy follows the protocol: Perception mainly requires recovering physical entities from the annotated first frame, Formulation requires abstracting or instantiating the governing law, and Deduction requires rolling that law forward into full temporal dynamics. Thus, video models do not fail uniformly; they often capture local visual or textual cues, but struggle once those cues must be converted into law-conditioned temporal evolution.

Subtrack-level failure modes. The subtracks expose finer-grained bottlenecks within each stage. In Perception, Perception-Text is generally easier than Perception-Graphic, suggesting that reading visible quantities is easier than grounding them to object regions. In Formulation, Formulation-Text and Formulation-Graphic reveal a gap between symbolic law selection and grounded state prediction: a model may choose a plausible equation without correctly placing objects or velocities at the target time. In Deduction, the challenge becomes fully temporal, requiring models to maintain object identity, velocity, acceleration, and contact events across frames; visually smooth motion can still violate the law-derived trajectory or collision response.

Reasoning funnel. Figure 4(b) summarizes this pattern as a reasoning funnel: average scores drop from Perception to Formulation and further to Deduction. Earlier success is therefore necessary but not sufficient. Good Perception does not guarantee correct Formulation, since copying annotations can remain superficial without identifying the governing law. Likewise, good Formulation does not guarantee correct Deduction, since selecting a law or predicting a single target state is easier than executing that law consistently over time. Unified models provide a useful reference, showing that explicit understanding modules help across stages but do not remove the deduction bottleneck.

Takeaway 2 Video models show a progressive reasoning bottleneck: they can often perceive physical cues and partially formulate laws, but these intermediate successes do not reliably transfer to law-consistent dynamics over time.

4.4 Pillar-wise and Source-wise Analysis

Generalization across laws. The pillar-wise results compare three single-law pillars, Grav., Mom., and N1, with the multi-law branch. As shown in Table 2 and Figure 4(c), Multi is generally harder than the single-law pillars. This gap reflects the compositional nature of multi-law cases: the state produced by one law, such as position, velocity, direction, or contact, must become the initial condition for the next. The lower Multi scores suggest that current video models can capture isolated law-specific motion priors, but remain weak at carrying physical states across law transitions, where state transfer and temporal consistency become critical.

Generalization across sources. The source-wise results reveal a complementary limitation. As shown in Figure 4(d), both video and unified models perform worse on real-world cases than on simulated cases, although unified models maintain higher absolute scores. Since the governing laws remain unchanged across sources, this Sim-to-Real drop mainly reflects failures in grounding and tracking under realistic visual conditions. Thus, stronger understanding modules help, but do not remove the need for robust generalization across visual domains.

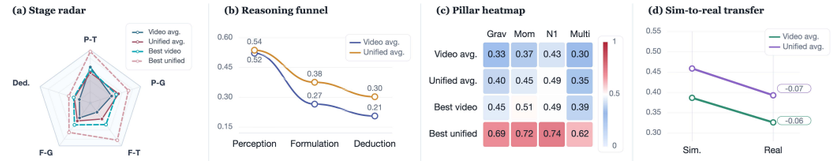


图 4：按阶段、支柱、来源分析。(a,b) 从感知到定律表述再到推演，得分下降。(c) 比较单、多定律。(d) 展示仿真到现实差距。

展示明确难度次序：感知最易，定律表述更难，推演最难。这与协议一致：感知主要恢复标注首帧的物理实体；定律表述抽象或实例化支配定律；推演将定律前向展开为完整时间动力学。因此视频模型并非统一失效，常能捕捉局部视觉或文字线索，但在线索须转为以定律为条件的时间演变时困难。

子赛道失效模式。子赛道揭示各阶段更细瓶颈。感知–文本通常比感知–图形容易，说明读取可见物理量比定位到物体区域容易。定律表述–文本与图形揭示符号定律选择和具场景依据的状态预测之间差距：模型可能选合理方程，却未在目标时间正确放置物体、速度。推演完全涉及时间，要求跨帧维持物体身份、速度、加速度和接触事件；视觉平滑运动仍可能违反定律推导轨迹或碰撞响应。

推理漏斗。图 4(b) 用漏斗概括：平均分从感知降到定律表述，再降到推演。因此前期成功必要但不充分。好的感知不保证正确定律表述，复制标注可能仅是表面行为，没有识别支配定律；好的定律表述也不保证正确推演，选择定律或预测单一目标状态比随时间一致执行容易。统一模型提供参考，显示显式理解模块有益于各阶段，却未消除推演瓶颈。

要点 2：视频模型呈递进推理瓶颈：常能感知物理线索、部分表述定律，但这些中间成功不能可靠转为随时间一致的动力学。

4.4 按支柱与来源分析

跨定律泛化。按支柱结果比较万有引力 Grav.、动量 Mom.、牛顿第一定律 N1 三个单定律支柱与多定律分支。表 2、图 4(c) 显示 Multi 通常更难。这反映组合性质：前一定律产生的位置、速度、方向或接触状态，须成为下一定律初始条件。较低 Multi 分数表明当前视频模型可捕捉孤立的定律运动先验，却不善于跨定律转移传递物理状态，而状态传递和时间一致性在此至关重要。

跨来源泛化。图 4(d) 显示视频与统一模型在真实案例均弱于仿真，尽管统一模型绝对分更高。各来源定律不变，因此仿真到现实下降主要反映真实视觉条件下定位、跟踪失效。更强理解模块有益，但不能消除跨视觉领域稳健泛化的需求。

Takeaway 3 Video models show limited generalization across both laws and sources: they struggle to carry physical states across law transitions in multi-law cases and to apply the same laws under real-world visual variation.

4.5 Qualitative Failure Analysis

To complement the quantitative results, Figure 5 visualizes representative failure cases. A common failure is annotation semantics rather than perception: the model may preserve the text and track the object, yet fail to bind an annotation to its physical meaning. For example, an initial-velocity arrow can be visually recognized but assigned the wrong direction as a physical initial condition, causing both the Formulation-Graphic target state and the Deduction trajectory to evolve incorrectly. This shows that copying annotations does not imply understanding the underlying physical variables.

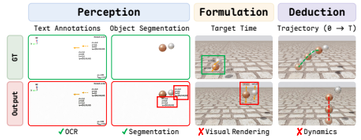


Figure 5: **Qualitative failure analysis.** The model preserves annotations and objects, but misbinds the initial-velocity cue, yielding a wrong target state and trajectory.

A second class of failures occurs before physics reasoning begins. Some models fail to follow the requested output format, such as blurring annotation text, failing to isolate target objects, producing unreadable formulas, or omitting velocity arrows and labels. These errors reflect limitations in OCR, segmentation, rendering, and instruction following.

4.6 Protocol Ablations

We ablate two protocol choices in Apple-π: how physical quantities are specified and how detailed the task prompt should be. Table 3 reports paired deltas against the default setting.

Text-Parameter. Providing physical quantities in a structured text prompt is nearly neutral: the average change is small for both models, and all available track-level deltas stay within ± 0.05. This indicates that Apple-π is not primarily bottlenecked by reading infographic annotations. We therefore keep infographic annotations not as a performance shortcut, but as a compact, spatially grounded, and auditable way to bind each physical quantity to its visual referent.

Table 3: **Protocol ablations.** Values are paired score deltas from the default setting; positive means better, and "—" marks inapplicable subtracks.

Model	Avg. Δ	Track-wise Δ				
		P-T	P-G	F-T	F-G	Ded.
— Text-Parameter —						
Veo 3.1 [63]	+0.008	—	+0.020	+0.038	-0.024	+0.000
Nano Banana 2 [19]	+0.020	—	+0.044	-0.006	+0.011	+0.030
— Concise-Prompt —						
Veo 3.1 [63]	-0.018	+0.015	-0.068	+0.051	-0.096	+0.006
Nano Banana 2 [19]	-0.037	-0.016	-0.085	+0.051	-0.089	-0.045

Concise-Prompt. A concise track-level prompt with less detailed output specifications slightly lowers the average score. The drop is concentrated on graphic-output tracks, especially P-G and F-G, suggesting that the full prompt helps enforce visual constraints. Meanwhile, F-T improves by +0.051 for both models, showing that concise instructions are sufficient for formula selection. Overall, the full prompt mainly standardizes the visual answer format rather than injecting physics solutions.

5 Conclusion

We introduced Apple-π, a law-grounded benchmark for evaluating whether video generation models can move beyond visual physical plausibility toward auditable physical intelligence. Built on Orchard, Apple-π anchors each case in explicit classical-mechanics laws and decomposes evaluation into Perception, Formulation, and Deduction. Experiments show that current video models acquire useful physical priors from large-scale video training, but remain far from reliable law-grounded simulators, especially when quantities must be formulated into laws, executed over time, composed across law transitions, and transferred to real-world domains. Stronger unified models suggest that future video world models will require not only better temporal generation, but also explicit understanding modules, richer physical-reasoning data, and reasoning-oriented post-training.

要点 3：视频模型跨定律、来源泛化有限：难以在多定律转移中传递物理状态，也难以在真实视觉变化下应用同一定律。

4.5 定性失效分析

为补充定量结果，图 5 展示代表性失效案例。常见失效在标注语义而非感知：模型可能保留文字并跟踪物体，却不能将标注绑定到物理含义。例如初速度箭头被视觉识别，却被错误赋予物理初始条件方向，导致定律表述–图形的目标状态及推演轨迹均错误演变。这说明复制标注不等于理解底层物理变量。

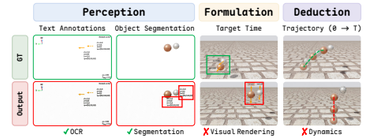


图 5：定性失效。模型保留标注和物体，却错误绑定初速度线索，产生错误目标状态及轨迹。

第二类失效发生在物理推
理开始之前。有些模型不遵循输出格式，如标注模糊、未隔离目标物体、公式不可读、遗漏速度箭头/标签。这反映 OCR、分割、渲染、指令遵循局限。

4.6 协议消融

我们消融 Apple-π 的两个协议选择：物理量如何指定、任务提示应多详细。表 3 报告相对默认设置的配对差值。

文本参数。用结构化文字提示提供物理量几乎无影响：两个模型平均变化小，全部可用赛道差值位于 ± 0.05 内。这说明 Apple-π 主要瓶颈并非读取信息图标注。保留信息图是因为它紧凑、具有空间依据、可审计，便于将各物理量绑定视觉对象，而非用作性能捷径。

表 3：协议消融。数值为相对默认的配对评分差；正值更好，“—”表示不适用。

模型	平均值 Δ	各赛道 Δ				
		P-T	P-G	F-T	F-G	Ded.
— 文本参数 —						
Veo 3.1 [63]	+0.008	—	+0.020	+0.038	-0.024	+0.000
Nano Banana 2 [19]	+0.020	—	+0.044	-0.006	+0.011	+0.030
— 简洁提示 —						
Veo 3.1 [63]	-0.018	+0.015	-0.068	+0.051	-0.096	+0.006
Nano Banana 2 [19]	-0.037	-0.016	-0.085	+0.051	-0.089	-0.045

简洁提示。减少输出细节的简洁赛道提示使平均分略降，下降集中在图形输出，尤其 P-G、F-G，表明完整提示帮助约束视觉输出。同时，两个模型 F-T 均提高 +0.051，说明简洁指令足以选择公式。总体上完整提示规范视觉答案格式，而非注入物理解。

5 结论

我们提出 Apple-π，基于定律的基准，检验视频生成模型能否超越视觉物理合理性，实现可审计物理智能。以 Orchard 为基础，Apple-π 将各案例锚定于显式经典力学定律，分解为感知、定律表述、推演评估。实验显示，当前视频模型从大规模视频训练获得有用物理先验，却远非可靠定律仿真器，尤其在物理量转为定律、随时间执行、跨定律转移组合及真实领域迁移时。更强统一模型提示未来视频世界模型不仅需更好时间生成，还需显式理解模块、更丰富物理推理数据及推理导向后训练。

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A Orchard Design Principles and Data Card

A.1 Design Principles

Law-first rather than video-first. Orchard is curated from a law-first perspective. Instead of starting from arbitrary videos and retrospectively assigning physical labels, each case begins with an explicit governing principle in classical mechanics, a specified initial condition, and a law-predicted future trajectory. This design makes Apple-π diagnostic: when a model fails, we can localize the failure to Perception, Formulation, or Deduction rather than treating the generated video as an undifferentiated endpoint.

Single-law diagnosis and multi-law generalization. The dataset is divided into a single-law branch and a multi-law branch. Single-law cases isolate one dominant physical principle, enabling clean diagnosis with minimal confounding. Multi-law cases compose two or more principles, such as an inclined-plane stage followed by projectile motion or a curved-track stage followed by collision. These cases test whether a model can transfer state variables from one law to the next: position, velocity, contact status, object identity, and direction of motion.

Controlled object vocabulary. Across simulated, self-recorded, and Internet-sourced cases, we standardize the dynamic objects to four primitive solids: sphere, cube, cylinder, and cone. This does not make the benchmark purely synthetic or visually trivial. Rather, it controls object-specific semantic priors and yields consistent centers, masks, contact surfaces, and size parameters, which are necessary for reproducible law-grounded evaluation.

Spatially grounded physical quantities. Physical quantities are attached to visual referents. A mass belongs to an object; a friction coefficient belongs to a contact surface; an angle belongs to a ramp; a velocity vector belongs to a moving body and direction. We therefore provide an infographic-style annotated first frame. The annotations specify initial conditions but not the answer: models must still identify the governing law and deduce the future state or trajectory. This interface reduces irrelevant reference-resolution burden while preserving the intended physical reasoning challenge.

A.2 Dataset Composition and Native Frame Rates

Orchard contains 400 cases spanning ten canonical mechanics tasks. The dataset combines three sources with different ground-truth construction procedures and native frame rates, summarized in Table 4. Importantly, simulator and real-world cases are not treated as identical: simulator cases provide programmatic ground truth from the physics engine, while real-world cases provide measurement-backed and human-reviewed annotations.

Table 4: Data sources and native ground-truth frame rates. Frame rate is stored per case and used by the evaluation pipeline; Apple-π does not assume a single global GT fps.

Source	Cases	Native GT fps	Ground-truth source
Simulated	243	24 fps	Isaac Sim/PhysX state, render passes, masks, depth, object pose, velocity, contact events, and analytically verified labels.
Self-recorded real	121	30 fps	Controlled laboratory recordings with instrumented measurements, human-refined masks, and physics-derived state annotations.
Internet-sourced real	36	Native fps of each video	Curated physics-demonstration clips with per-video fps stored in metadata; only clips with auditable motion windows and sufficient annotation information are retained.

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A Orchard 设计原则与数据卡

A.1 设计原则

定律优先而非视频优先。Orchard 从定律视角构建；不是先任意选视频再追溯标记物理标签，而是每例从显式经典力学原理、规定初始条件与定律预测未来轨迹出发。这使 Apple-π 可诊断：失效可定位到感知、定律表述或推演，而非将生成视频视为未区分的端点。

单定律诊断与多定律泛化。数据分单、多定律分支。单定律隔离一个主导物理原理，以最小混杂进行清晰诊断。多定律组合两个以上原理，如斜面后抛体或曲轨后碰撞，检验模型能否将位置、速度、接触状态、物体身份、运动方向从前一定律传至下一定律。

受控物体词汇。仿真、自录、互联网案例均将动态物体统一为球、立方体、圆柱、圆锥四种基本实体。这不意味着基准纯合成或视觉简单，而是控制物体语义先验，提供可复现定律评估所需的一致中心、掩码、接触面及尺寸参数。

具有空间依据的物理量。物理量附着于视觉对象：质量属于物体，摩擦系数属于接触面，角度属于斜坡，速度向量属于运动体及方向。因此提供信息图式标注首帧。标注给出初始条件而非答案：模型仍须识别支配定律、推演未来状态或轨迹。这减少无关指代解析负担，保留预期物理推理挑战。

A.2 数据集构成与原生帧率

Orchard 含 400 例，跨十个典型力学任务。数据结合三种来源，其真值构建流程与原生帧率不同，见表 4。仿真与真实案例不视为相同：前者由引擎给出程序真值，后者给出测量支撑、人工审核的标注。

表 4：数据来源与真值原生帧率。帧率按案例存储并用于评估；Apple-π 不假设统一全局真值 fps。

来源	样例	真实值原始帧率	真实值来源
模拟	243	24 帧/秒	Isaac Sim/PhysX 状态、渲染通道、掩码、深度、物体姿态、速度、接触事件与解析验证的标签。
自行录制的真实视频	121	30 帧/秒	受控实验室记录，配有仪器实测数据，人工修正的掩码，与物理推导状态标注。
来自互联网的真实视频	36	各视频的原始帧率	精选物理演示片段，各视频帧率存于元数据；仅保留具有可审计运动窗口和充分标注信息的片段。

Table 5: Task taxonomy in Orchard. The first nine tasks are single-law diagnostic tasks; the last row contains multi-law compositions.

Task	Pillar	Representative law	Evaluated physical state
Free fall	Universal gravitation	$v(t) = v_0 + gt, h(t) = h_0 + v_0 t + \frac{1}{2} g t^2$	Vertical position and velocity.
Projectile motion	Universal gravitation	$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{g} t^2$	2D/3D trajectory and velocity direction.
Inclined plane	Universal gravitation	$a = g(\sin\theta - \mu\cos\theta)$, with $\mu = 0$ for frictionless cases	Ramp-aligned acceleration and object position.
Circular motion	Universal gravitation	$mg \approx mv^2/r$ or energy-derived speed along the track	Track following, tangent velocity, and contact.
Perfectly elastic collision	Conservation of momentum	Momentum conservation with $e = 1$	Post-collision velocity and identity preservation.
Perfectly inelastic collision	Conservation of momentum	Momentum conservation with $e = 0$	Merged motion and shared velocity.
Inelastic collision	Conservation of momentum	Momentum conservation with $0 < e < 1$	Partially dissipative collision response.
At rest	Newton's first law	$\sum \mathbf{F} = 0,\ \mathbf{v} = 0$	Static object and absence of drift/deformation.
Uniform linear motion	Newton's first law	$\sum \mathbf{F} = 0,\ \mathbf{v}(t) = \mathbf{v}_0$	Constant-speed straight-line motion.
Composition	Multi-law	Sequential composition of the above laws	State transfer across law transitions.

Table 6: Canonical case-level metadata and artifacts.

Artifact	Description
`first_frame.png`	Unannotated first frame, when available.
`annotated_first_frame.png`	Infographic-style input frame used for all five subtracks.
`gt/video.mp4` or `gt/rgb/`	Ground-truth video or RGB frame directory.
`gt/masks/`	Per-frame object masks for objective IoU and masked metrics.
`case.json`	Source type, task type, object list, native GT fps, dimensions, camera setting, and source notes.
`physics_duration.txt`	Requested physical duration T in seconds.
`timestamps.json`	Canonical evaluation timestamps $\mathcal{T}_c$.
`formula_info.json`	Four-way formula candidates, correct option, symbolic formula, and substituted values.
`velocity.json`	Target time $t^\star$, object centers, velocity vectors, speeds, and velocity-label metadata.
`measurement.json`	Real-world measurement records, when applicable.

A.3 Task Taxonomy

Table 5 lists the task families. The formula column gives the representative law used to construct or verify the case; individual instances may include task-specific parameters such as non-zero initial velocity, friction, restitution, or non-Earth gravitational acceleration.

A.4 Released Metadata

Each case is released as a self-contained directory. Table 6 summarizes the canonical files. Source-specific fields are optional, but the evaluation-critical fields are always present: native GT fps, physical duration, timestamp list, object masks when available, formula metadata, and target-time state.

表 5：Orchard 任务分类。前九个单定律诊断任务；最后一行为多定律组合。

Task	支柱	代表性定律	评估的物理状态
自由落体	万有引力	$v(t) = v_0 + gt, h(t) = h_0 + v_0 t + \frac{1}{2} g t^2$	竖直位置与速度 ity.
抛体运动	万有引力	$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{g} t^2$	二维/三维轨迹与速度方向。
斜面	万有引力	$a = g(\sin\theta - \mu\cos\theta)$，其中 $\mu = 0$ 用于无摩擦案例	沿斜面的加速度及物体位置。
圆周运动	万有引力	$mg \approx mv^2/r$ 或能量推算的速率	沿轨道、切向速度与接触。
完全弹性碰撞	动量守恒	沿赛道 动量 守恒 其中 $e = 1$	碰撞后速度与身份保持。
sion	tum		
完全非弹性碰撞	动量守恒	动量 守恒 其中 $e = 0$	合并后的运动与共同速度。
碰撞	tum		
非弹性碰撞	动量守恒	动量 守恒 其中 $0 < e < 1$	部分 耗散碰撞碰撞响应。
静止	牛顿第一定律	$\sum \mathbf{F} = 0, \mathbf{v} = 0$	静止物体，无漂移/变形。
匀速直线运动	牛顿第一定律	$\sum \mathbf{F} = 0, \mathbf{v}(t) = \mathbf{v}_0$	匀速直线运动。
组合	多定律	顺序组合 上述定律	跨定律的状态传递转换。

表 6：规范案例元数据与文件。

文件	描述
first_frame.png	无标注首帧（可用时）。
annotated_first_frame.png	五子赛道共同使用的信息图式输入帧。
gt/video.mp4 or gt/rgb/	真值视频或 RGB 帧目录。
gt/masks/	用于客观 IoU 与掩码指标的逐帧物体掩码。
case.json	来源类型、任务类型、物体列表、真值原生 fps、尺寸、相机设置与来源说明。
physics_duration.txt	请求物理时长 T，单位秒。
timestamps.json	规范评估时间戳 $\mathcal{T}_c$。
formula_info.json	四候选公式、正确选项、符号公式及代入值。
velocity.json	目标时刻 $t^\star$、物体中心、速度向量、速率和速度标注元数据。
measurement.json	真实测量记录（适用时）。

A.3 任务分类

表 5 列任务族。公式列为构建或验证案例的代表定律；实例可含非零初速度、摩擦、恢复系数、非地球重力加速度等专用参数。

A.4 发布元数据

每例以独立目录发布，规范文件见表 6。来源特定字段可选，但评估关键字段总是存在：真值原生 fps、物理时长、时间戳列表、可用物体掩码、公式元数据、目标时刻状态。

Table 7: Real-world physical-parameter measurement. The velocity procedure uses the recorded fps: 30 fps for self-recorded clips and the native fps for Internet-sourced clips.

Quantity	Instrument or procedure	Precision / convention
Object mass	Digital balance	±0.01 g
Object size	Digital caliper	±0.05 mm
Drop / launch height	Steel tape plus frame-interpolation check	±1 mm
Initial velocity	Frame-by-frame velocimetry at recorded fps	±0.05 m/s
Coefficient of restitution	Two-drop ratio test on the surface	±0.02
Friction coefficient	Inclined-plane tilt-threshold test	±0.05
Gravitational acceleration	Case metadata; default Earth value for ordinary real-world clips	Stored per case

B Ground Truth, Annotation, and Quality Assurance

B.1 Simulator Ground Truth

Simulator cases are generated in Isaac Sim with deterministic case configurations. Each configuration specifies dynamic objects, primitive shape, material, geometry, initial pose, initial linear/angular velocity, environment parameters, ground properties, camera pose, and rendering options. The simulator subset is programmatic-first: RGB frames, masks, depth, object poses, velocities, and contact events are read from the engine state or render passes rather than manually estimated.

Two consistency rules are used during case authoring. First, mass is derived from material density and analytic primitive volume, $m = \rho V$, so changing a shape or material updates the physical mass consistently. Second, visual material and physics material share a single material key, preventing a case from having a visually metallic object with unrelated density, friction, or restitution parameters. When a task admits a closed-form solution, such as ideal collision velocities or projectile states, the analytic value is used as the canonical answer key and the engine state is retained for diagnostics.

B.2 Real-World Ground Truth

Real-world cases are measurement-first rather than engine-state-first. Self-recorded clips are captured at 30 fps under controlled laboratory conditions. Internet-sourced clips retain their native fps, which is stored in the case metadata. For both real-world sources, physical parameters are obtained through instrumented measurements, visible calibration cues, public object specifications when appropriate, and human review. The goal is not to make real videos identical to simulated videos, but to keep the governing law auditable under realistic appearance, lighting, texture, and camera conditions.

B.3 Mask and Tracking Annotation

Object masks are generated with a semi-automatic pipeline. Initial masks are produced by a segmentation model, propagated through the clip, and then inspected by human annotators. If the automatic masks show drift, over-segmentation, under-segmentation, or identity switches, annotators correct the mask at the failing frame and re-run propagation. For multi-object cases, object identity is checked across the full evaluation window, with special attention to contact and collision frames.

For simulator cases, engine-rendered instance masks are canonical. For real-world cases, the human-refined masks are canonical. These masks support both graphic-subtrack IoU and Deduction trajectory metrics.

B.4 Formula Answer-Key Construction

For each case, `formula_info.json` contains four candidate formulas:

1. the correct governing formula for the case;
2. a confusing real formula that shares symbols with the annotations but does not govern the case;

表 7：真实物理参数测量。速度流程使用记录帧率：自录 30 fps，互联网使用原生 fps。

物理量	仪器或测量过程	精度／约定
物体质量	电子天平	±0.01 g
物体尺寸	数显卡尺	±0.05 mm
下落／发射高度	钢卷尺与帧插值检查	±1 mm
初始速度	在记录帧率下逐帧测速 fps	±0.05 m/s
恢复系数	表面两次下落比值测试	±0.02
摩擦系数	斜面倾斜临界角测试	±0.05
重力加速度	案例元数据；默认地球数值用于普通现实视频片段	按样例存储

B 真值、标注与质量保证

B.1 仿真器真值

Isaac Sim 以确定性案例配置生成仿真，规定动态物体、基本形状、材料、几何、初姿态、初线/角速度、环境参数、地面属性、相机姿态和渲染选项。仿真子集优先程序化：RGB、掩码、深度、姿态、速度、接触事件从引擎状态或渲染通道读取，而非人工估计。

案例构建使用两条一致性规则。首先，质量由材料密度和基本实体解析体积 $m = \rho V$ 推导，形状、材料改变时质量同步更新。其次，视觉材质与物理材料共享一个材料键，避免视觉金属物体对应无关密度、摩擦、恢复系数。任务存在闭式解时，如理想碰撞速度或抛体状态，以解析值作为规范答案键，保留引擎状态供诊断。

B.2 真实世界真值

真实案例优先使用测量，而非引擎状态。自录片段在受控实验室以 30 fps 拍摄；互联网片段保留原生帧率并存入元数据。两种真实来源均通过仪器测量、可见校准线索、适用的公开物体规格及人工审核获得物理参数。目标不是让真实视频与仿真完全相同，而是在真实外观、照明、纹理、相机条件下保持定律可审计。

B.3 掩码与跟踪标注

物体掩码以半自动流程生成：分割模型生成初始掩码，传播至全片，再由标注员检查。自动掩码若有漂移、过分割、欠分割或身份交换，人工修正失败帧并重新传播。多物体案例检查完整评估窗口的身份一致性，特别关注接触、碰撞帧。

仿真采用引擎实例掩码为规范真值，真实采用人工修正掩码。二者支持图形子赛道 IoU 与推演轨迹指标。

B.4 公式答案键构建

每例 formula_info.json 含四候选公式：

1. 案例正确支配公式；2. 共享标注符号却不支配案例的易混淆真实公式；

Table 8: Examples of issues caught and corrected during simulator QA.

Fix category	Cases caught
Physics-duration adjustment	23
Velocity-label theoretical-value override	10
Configuration re-recording due to camera, material, or position issue	30

Table 9: Inter-annotator agreement on categorical fields.

Field	Cohen's κ
Object naming convention compliance	0.97
Task-type classification	1.00
Formula correct-answer choice	0.96
Camera-framing acceptability	0.91
Velocity-label legibility	0.89

3. an unrelated real formula from another mechanics family;
4. a fabricated formula that is syntactically plausible but not a valid physical law.

This construction distinguishes law selection from superficial symbol matching. A model that chooses a formula merely because it contains the same variables should fail on the confusing distractor.

B.5 Three-Pass Review Protocol

Every case undergoes a 1+2 review protocol: one primary annotation pass followed by two independent secondary review passes. Reviewers check the initial frame, annotation legibility, physical duration, target time, mask identity, velocity labels, formula candidates, and source metadata. Disagreements are resolved in the following order: analytic physical backstop when available; senior arbitration for visual or measurement ambiguities; domain-expert escalation for formula-choice ambiguity.

Table 8 reports issues caught by a complete simulator-subset review pass. These corrections indicate that the review protocol is not merely procedural but actively improves label quality.

B.6 Inter-Annotator Agreement

We conduct a retroactive inter-annotator agreement study across simulator and real-world categories. Two production annotators independently re-annotate held-out cases under blind conditions. Categorical fields are measured with Cohen's κ; continuous fields are measured with ICC(2,1) after quantization to the protocol grid or instrument precision. Results are shown in Tables 9 and 10.

C Benchmark Protocol and Prompt Templates

C.1 Five Subtracks

All subtracks receive the same annotated first frame and a subtrack-specific chain-of-frames prompt. The expected output differs by subtrack, as summarized in Table 11.

C.2 Prompt Families

We use two prompt families because evaluated models expose different interfaces. Unified understanding-generation models produce still images for the four non-Deduction subtracks and timestamped keyframes for Deduction. Video-generation models produce an MP4 for every subtrack; for the four non-Deduction subtracks, the last frame of the generated video is used as the answer artifact.

表 8：仿真质量检查发现并修正的问题示例。

修正类别	发现的样例
物理时长调整	23
速度标签理论值覆盖修正	10
因相机、材料或位置重新记录配置问题	30

表 9：类别字段标注员间一致性。

字段	Cohen κ
是否遵循物体命名规范	0.97
任务类型分类	1.00
公式正确选项	0.96
相机取景可接受性	0.91
速度标注的可辨识性	0.89

3. 另一力学族的无关真实公式；
4. 语法合理却非有效物理定律的虚构公式。

此构造区分定律选择与表面符号匹配。仅因公式含相同变量而选择的模型，应在易混淆干扰项上失败。

B.5 三轮审核协议

每例经过 1+2 审核协议：一轮主标注及两轮独立复核。检查首帧、标注可读性、物理时长、目标时刻、掩码身份、速度标签、公式选项、来源元数据。分歧按顺序解决：可用时优先解析物理依据；视觉或测量歧义由资深人员裁决；公式选择歧义上报领域专家。

表 8 报告完整仿真子集审核发现的问题。修正说明审核不仅是程序规定，也实际改善标签质量。

B.6 标注员间一致性

我们对仿真与真实类别开展追溯标注员一致性研究。两名生产标注员在盲条件下独立重标保留案例。类别字段以 Cohen's κ 衡量；连续字段量化至协议网格或仪器精度后以 ICC(2,1) 衡量。结果见表 9、10。

C 基准协议与提示模板

C.1 五个子赛道

全部子赛道接收相同标注首帧及专用帧链提示。预期输出按赛道不同，见表 11。

C.2 提示族

因模型接口不同，使用两类提示。统一理解–生成模型在四非推演赛道输出静态图，推演输出时间戳关键帧；视频模型各赛道均输出 MP4，四非推演赛道取末帧作答案。

Table 10: Inter-annotator agreement on continuous fields.

Field	ICC(2,1)
Physics duration	0.94
Target time	0.96
Real-world object mass	0.99
Real-world drop / launch height	0.98
Real-world coefficient of restitution	0.88
Real-world initial velocity	0.95

Table 11: Input-output specification for the five Apple-π subtracks.

Subtrack	Question	Expected output	Main evaluation target
P-T	Read physical quantities	Final white-background annotation artifact	Text/OCR, annotation placement, readability
P-G	Ground physical objects	Final white-background object-only artifact	Object localization and mask IoU
F-T	Select governing law	Final white-background three-line formula answer	Option, formula, substitution
F-G	Predict state at $t^\star$	Scene at target time with velocity arrows and labels	Target-state mask IoU, arrow direction, speed label
Deduction	Generate full dynamics	Full trajectory video or timestamped keyframes	Law-consistent temporal evolution

Runtime placeholders include `{formula_choices}`, `{target_time}`, `{time_point}`, and `{physics_duration}`. The prompts below are reproduced from the released code's machine-readable templates, `UM_PROMPT_TEMPLATES` and `VM_PROMPT_TEMPLATES`; source em dashes are rendered as - for `listings` compatibility under pdfLaTeX.

C.2.1 Unified Understanding-Generation Models

Unified models receive a single still-image directive per non-Deduction subtrack. For the Deduction subtrack, they receive the per-keyframe directive below once per requested timestamp.

Perception-Text.

```
Given this physics infographic, generate a new image that shows ONLY the human-drawn
  ↪ annotations on a pure white background. Keep all text labels, coordinate axes, and
  ↪ velocity arrows exactly as they appear -- same position, content, color, font, and size.
  ↪  Text must be crisp and readable. Remove everything else: the background, ground, sky,
  ↪ and all physical objects. Do not add any extra elements not in the original.
```

Perception-Graphic.

```
Show only the subject physical objects from this image on a pure white background. Remove
  ↪ all annotations (text labels, coordinate axes, velocity arrows), the ground surface,
  ↪ sky, and supporting structures (ramps, tracks, rails, platforms). Keep only the objects
  ↪  that have labeled physical properties. Do not transform, resize, or replace any object
  ↪  -- a cube must stay a cube, a sphere must stay a sphere. Each object must retain its
  ↪ exact position, size, shape, color, and texture.
```

Formulation-Text. The runtime placeholder `{formula_choices}` is filled with the case's four candidate formulas read from `formula_info.json`.

```
This physics infographic shows a scene with annotated physical parameters. Below are 4
  ↪ candidate formulas. Exactly ONE is the correct governing equation for the motion shown:
{formula_choices}

Generate an image with a pure white background containing THREE lines of large, clearly
  ↪ readable, centered text:
  Line 1: The correct option letter (A/B/C/D).
```


表 10：连续字段标注员间一致性。

字段	ICC(2,1)
物理时长	0.94
目标时刻	0.96
真实物体质量	0.99
真实下落 / 发射高度	0.98
真实恢复系数	0.88
真实初始速度	0.95

表 11：Apple-π 五子赛道的输入输出规格。

子赛道	问题	预期输出	主要评估目标
P-T	读取物理量	最终白底标注文件	文本/OCR、标注位置、可读性
P-G	定位物理物体	最终白底物体唯一答案文件	物体定位与掩码 IoU
F-T	选择支配运动的定律	白底最终三行公式答案	选项、公式、代入恢复
F-G	预测 $t^\star$ 时状态	目标时刻场景，含速度箭头与标签	目标状态掩码 IoU、箭头方向、速率标签
推演	生成完整动力学过程	完整轨迹视频或时间戳带时间戳的关键帧	符合定律的时间演化

运行时占位符有 {formula_choices}、{target_time}、{time_point}、{physics_duration}。下列提示摘自发布代码的机器可读模板 UM_PROMPT_TEMPLATES、VM_PROMPT_TEMPLATES；为兼容 pdfLaTeX 代码列表，源长破折号渲染为 –。

C.2.1 统一理解–生成模型

统一模型每非推演赛道接收一个静态图指令；推演则每请求时间戳接收一次下列关键帧指令。

感知–文本。

```
给定物理信息图，生成新图，只在纯白背景显示人工绘制的↪ 标注。全部文本、坐标轴及↪ 速度箭头保持
原样：位置、内容、颜色、字体、大小均相同。↪ 文字清晰可读。移除其余一切：背景、地面、天空、↪ 全
部物体。不增加原图不存在的元素。
```

感知—图像。

```
在纯白背景上只显示图中主体物体。移除↪ 全部标注（文本、坐标轴、速度箭头）、地面、↪ 天空及斜坡、
轨道、导轨、平台等支撑结构。仅保留↪ 带物理属性标签的物体。不变换、缩放或替换物体↪ ：立方体仍
是立方体，球仍是球。各物体严格保持↪ 位置、大小、形状、颜色、纹理。
```

定律表述–文本。{formula_choices} 从 formula_info.json 读取四候选填充。

```
该物理信息图含标注参数。下有四个↪ 候选公式，恰一个是所示运动正确主导方程：{formula_choices}
```

```
生成纯白背景图，含三行大号、清晰↪ 可读、居中文字。第 1 行：正确选项字母 A/B/C/D。
```

```
  Line 2: The correct formula exactly as written above.
  Line 3: The formula with annotated variables replaced by their numeric values from the
  ↪ image, keeping unannotated symbols unchanged.

When substituting: if multiple objects exist, pick exactly one consistent set of values.
```

Formulation-Graphic. The runtime placeholder `{target_time}` is the case-specific target instant $t^\star$.

```
Predict the physical state at t = {target_time} s. Generate an image showing the scene at
  ↪ that instant, where the original annotations (text labels, coordinate axes, arrows)
  ↪ have been cleanly removed while the background, ground, sky, and all scene lighting
  ↪ remain completely unchanged from the input. Show each object at its correct position at
  ↪  t = {target_time} s -- each object must retain its exact texture, material, size, and
  ↪ shape. For each moving object, overlay a borderless orange arrow starting from the
  ↪ object's center pointing in its instantaneous velocity direction, with a label "v = X.
  ↪ XX" (speed value, no unit). If an object is at rest, show "v = 0" with no arrow. Keep
  ↪ the same viewpoint as the input image -- no zoom, no rotation, no crop.
```

Deduction. The runtime placeholder `{time_point}` is filled with the requested timestamp for each keyframe.

```
This image is an annotated physics scene. All annotations follow SI units: g is
  ↪ gravitational acceleration, e is restitution (0-1), v0 is initial velocity (arrows show
  ↪  direction), mu is friction coefficient. Generate what the scene looks like at t = {
  ↪ time_point} seconds after motion begins, following these physical parameters accurately.
  ↪  Remove all annotations (text, axes, arrows). Objects are rigid bodies -- same shape,
  ↪ size, color, texture throughout. Static camera, fixed background and ground.
```

C.2.2 Video-Generation Models

Video-generation models receive a chain-of-frames directive that elicits one video. For the four non-Deduction subtracks, the prompt asks the video to settle on the answer artifact and the last frame is used for scoring; for Deduction, the full generated video is scored.

Perception-Text.

```
Generate a video where the background, ground, sky, and all physical objects gradually fade
  ↪ to pure white, while all human-drawn annotations (text labels, coordinate axes,
  ↪ velocity arrows) remain completely unchanged. Static camera, no zoom, no rotation. The
  ↪ final frame should show only the original annotations on white -- crisp, sharp, and
  ↪ readable. Do not add any extra annotations or elements not in the original.
```

Perception-Graphic.

```
Generate a video where the background, ground, sky, and all annotations (text labels,
  ↪ coordinate axes, velocity arrows) gradually fade to pure white, while the subject
  ↪ physical objects remain still and unchanged. Static camera, no zoom, no rotation. The
  ↪ final frame should show only the objects with labeled physical properties on white --
  ↪ remove supporting structures (ramps, tracks, rails, platforms). Do not transform,
  ↪ resize, or replace any object. Each object must retain its exact position, size, shape,
  ↪  color, and texture.
```

Formulation-Text. As in the unified-model variant, `{formula_choices}` is filled from `formula_info.json`.

```
This physics infographic shows annotated physical parameters describing a specific type of
  ↪ motion. Below are 4 candidate formulas. Exactly ONE is correct:
{formula_choices}

Generate a video ending on a white background with THREE lines of large, clearly readable,
  ↪ centered text:
  Line 1: the correct option letter (A/B/C/D).
  Line 2: the correct formula exactly as written above.
  Line 3: the formula with annotated values substituted in.
```



```
  第 2 行：上文原样正确公式。第 3 行：将标注变量代入 ↪ 图像数值，未标注符号保持不变。

代入时，多物体须恰选一组一致数值。
```

For定律表述–图形。{target_time} 为案例目标时刻。 t $t^\star$。

```
Pr预测秒 = 此物理状态 s。生成一张图像，显示该时刻 ↪ 场景的图像。干净移除原标注（文本、坐标轴、箭头）↪ ，背景、地面、天
  空及全部照明 ↪ 与输入完全一致。各物体位于 t = {target_time} 秒的正确位置 ↪ ，严格保持纹理、材
  质、尺寸与 ↪ 形状。对运动物体，从中心叠加无边框橙色箭头 ↪ ，指向瞬时速度方向，标签为“v = X.↪ XX”（速度数值，无单位）。静止物体显示“v = 0”，不画箭头。保持 ↪ 输入视角，不缩放、不旋转、不裁剪。
```

推演。{time_point} 填入每关键帧请求时间戳。

```
图像是有标注的物理场景。全部采用 SI 单位：g 为 ↪ 重力加速度，e 为恢复系数（0–1），v0 为初速度（箭头示 ↪ 方向），mu 为摩擦系数。准确遵循参数，生成运动开始后 t = {↪ time_point} 秒的场景。↪ 移除全部标注（文字、轴、箭头）。物体为刚体，始终保持形状、↪ 大小、颜色、纹理。固定相机、背景、地面。
```

C.2.2 视频生成模型

视频模型接收诱导单视频的帧链指令。四非推演赛道要求视频最终停留在答案，末帧用于评分；推演评分完整视频。

感知–文本。

```
生成视频，让背景、地面、天空及全部物体逐渐淡 ↪ 为纯白，同时所有人工标注（文本、坐标轴、↪ 速度箭头）完全不变。固定相机，不缩放、不旋转。↪ 最终帧仅在白底显示原标注，清晰、锐利且 ↪ 可读。不添加原图没有的标注或元素。
```

感知—图像。

```
G生成视频，让背景、地面、天空及全部标注（文本、↪ 坐标轴、速度箭头）逐渐淡为纯白，同时主体 ↪ 物体保持静止不变。固定相机，不缩放、不旋转。↪ 最终帧只在白底显示带物理属性标签的物体；↪ 移除斜坡、轨道、导轨、平台等支撑结构。不变换、↪ 缩放或替换物体。各物体严格保留位置、尺寸、形状、↪ 颜色与纹理。
```

定律表述–文本。与统一模型一样，{formula_choices} 来自 formula_info.json。

```
信息图标注描述特定 ↪ 运动的物理参数。四候选公式中恰一正确：{formula_choices}
```

```
生成以白底结束的视频，含三行大号、清晰可读、↪ 居中文字：第 1 行，正确选项字母 A/B/C/D；第 2 行，与上文完全相同的正确公式；第 3 行，代入标注数值的公式。
```

Table 12: Canonical evaluation packets for different model families.

Subtrack	Unified models	Video-generation models
P-T / P-G / F-T / F-G	Generated PNG, evaluated directly as the answer artifact.	Generated MP4 decoded; last frame used as the answer artifact.
Deduction	Timestamped keyframes generated at $\mathcal{T}_c$.	Generated MP4 decoded; frames sampled after time-rescaling to the requested physical duration.

```
When substituting: if multiple objects exist, pick one consistent set.
```

Formulation-Graphic. As in the unified-model variant, `{target_time}` is the case-specific target instant.

```
Predict the physical state at t = {target_time} s. Generate a video where the original
  ↪ annotations (text labels, coordinate axes, arrows) gradually fade out and cleanly
  ↪ disappear, while the background, ground, sky, and all scene lighting remain completely
  ↪ unchanged throughout. Objects move smoothly to their correct positions at t = {
  ↪ target_time} s -- each object must retain its exact texture, material, size, and shape.
  ↪  For each moving object, draw a borderless orange arrow starting from the object's
  ↪ center pointing in its velocity direction, with a label "v = X.XX" (speed value, no
  ↪ unit). If at rest, show "v = 0" with no arrow. Static camera, no zoom, no rotation.
```

Deduction. The runtime placeholder `{physics_duration}` is the case's physics duration in seconds.

```
The first frame is an annotated physics scene. All annotations follow SI units: g is
  ↪ gravitational acceleration, e is restitution (0-1), v0 is initial velocity (arrows show
  ↪  direction), mu is friction coefficient. Generate {physics_duration} seconds of
  ↪ physically accurate motion strictly following these parameters. Every object AND the
  ↪ ground surface are perfectly rigid -- nothing breaks, cracks, deforms, or creates
  ↪ craters. Each object maintains its exact material, color, and texture throughout. All
  ↪ annotations must cleanly disappear after the first frame. Static camera, fixed
  ↪ background. Smooth, continuous motion.
```

C.3 Protocol Ablations

We use two ablations to verify that the protocol does not artificially inject the solution. In the Text-Parameter ablation, physical quantities are supplied in structured text rather than as infographic overlays. The small average change indicates that infographic annotations are not a shortcut, but a spatially grounded way to bind quantities to visual referents. In the Concise-Prompt ablation, we remove detailed output-format instructions. The main degradation appears on graphic-output tracks, showing that the full prompt mainly standardizes answer format rather than providing physics solutions.

D Evaluation Standardization Across Model Outputs

D.1 Output Canonicalization

Different models return different output formats, resolutions, fps values, and durations. Before scoring, every response is converted into a canonical evaluation packet, as shown in Table 12.

D.2 Time and FPS Alignment

Each case c defines a requested physical duration T_c and a native GT frame rate f_c^{GT}. The GT frame rate is source-dependent: 24 fps for simulator cases, 30 fps for self-recorded real-world cases, and the native fps stored in metadata for Internet-sourced cases.

表 12：不同模型族的规范评估包。

子赛道	统一模型	视频生成模型
P-T / P-G / F-T / F-G	生成的 PNG，直接评估作为答案文件。	解码 MP4；末帧作为答案。
推演	生成带时间戳的关键帧在 $\mathcal{T}_c$。	解码 MP4；各帧时间缩放后采样至要求的物理时长。

代入时，多物体选一组一致数值。

定律表述–图形。与统一模型一样，{target_time} 为案例目标时刻。

预测 t ={target_time} 秒的物理状态。生成视频，让原标注 ↪ （文本、坐标轴、箭头）逐渐淡出并干净消失 ↪ ，而背景、地面、天空及全部照明始终 ↪ 不变。物体平滑移动到 t = {↪ target_time} 秒的正确位置，严格保持纹理、材质、大小、形状。↪ 对每个运动物体，从中心画无边框橙色箭头 ↪ ，指向速度方向，标签“v = X.XX”（速度数值，无 ↪ 单位）。静止则显示“v = 0”，不画箭头。固定相机，不缩放、不旋转。

推演。{physics_duration} 为案例物理时长，单位秒。

首帧是有标注的物理场景。全部标注采用 SI 单位：g 为 ↪ 重力加速度，e 为恢复系数（0–1），v0 为初速度（箭头示 ↪ 方向），mu 为摩擦系数。生成 {physics_duration} 秒 ↪ 严格遵循参数的准确物理运动。所有物体及 ↪ 地面均为完全刚体，不破裂、开裂、变形或产生 ↪ 凹坑。各物体始终保持精确材质、颜色、纹理。全部 ↪ 标注在首帧后干净消失。相机静止，背景 ↪ 固定。运动平滑、连续。

C.3 协议消融

我们用两项消融验证协议未人为注入解答。文本参数消融用结构化文字替代信息图叠加提供物理量；平均变化小，说明信息图标注不是捷径，而是将物理量绑定视觉对象的空间化方式。简洁提示消融移除详细输出格式指令；下降主要在图形输出赛道，说明完整提示主要规范答案格式，而不提供物理解。

D 不同模型输出的评估标准化

D.1 输出规范化

模型返回不同格式、分辨率、fps、时长。评分前全部转换为规范评估包，见表 12。

D.2 时间与 FPS 对齐

每例 c 定义请求物理时长 T_c 与真值原生帧率 f_c^{GT}。帧率依来源：仿真 24 fps，自录真实 30 fps，互联网真实使用元数据原生 fps。

For a GT frame sequence with N_c^{GT} decoded frames, the GT frame nearest to physical time $t \in [0, T_c]$ is

$$i_c(t) = \mathrm{clip}\big(\mathrm{round}(t f_c^{\mathrm{GT}}),\ 0,\ N_c^{\mathrm{GT}} - 1\big)\,. \tag{1}$$

For a generated video, we do not interpret the provider's raw output duration as physical time. The prompt explicitly requests T_c seconds of motion, so the decoded generated sequence is treated as a continuous trajectory over $[0, T_c]$, regardless of whether the provider returns a nominal 5-second, 8-second, or other-length video. If the generated video contains N^{gen} frames, its effective evaluation fps is

$$\widehat{f}^{\mathrm{gen}} = \frac{N^{\mathrm{gen}}}{T_c}. \tag{2}$$

The generated frame nearest to physical time t is then

$$j_c(t) = \mathrm{clip}\Big(\mathrm{round}(t\widehat{f}^{\mathrm{gen}}),\ 0,\ N^{\mathrm{gen}} - 1\Big)\,. \tag{3}$$

This rule makes fps and duration comparable across models. Raw generated fps is used only for decoding; physical time is defined by the prompt and the case metadata.

Example. Suppose the GT clip is 4 seconds at 30 fps, so it has 120 frames. A model returns an 8-second video at 24 fps, so it has 192 decoded frames. Because the prompt requested "Generate 4 seconds of motion," we interpret the 192 generated frames as a 4-second trajectory and set $\widehat{f}^{\mathrm{gen}} = 192/4 = 48$ fps. At $t = 2$ seconds, we compare GT frame $i(2) = 60$ with generated frame $j(2) = 96$. The model's raw 8-second container duration is ignored for physical-time alignment.

D.3 Timestamp Grid

Each case stores a canonical timestamp set $\mathcal{T}_c \subset [0, T_c]$. Deduction metrics compare generated and GT frames at the same physical timestamps. For unified models, the timestamps in $\mathcal{T}_c$ are the keyframes requested from the model. For video-generation models, the video is decoded densely and sampled at the nearest generated frame $j_c(t)$ for each $t \in \mathcal{T}_c$. When event-critical moments exist, such as collision contact or ramp exit, they are included in $\mathcal{T}_c$ in addition to the uniform sampling grid.

D.4 Resolution Normalization

All pixel-level and mask-level metrics are computed at the GT resolution. If a generated image or frame has size $H_{\mathrm{gen}} \times W_{\mathrm{gen}}$ and the GT frame has size $H_{\mathrm{GT}} \times W_{\mathrm{GT}}$, the generated output is directly resized to $H_{\mathrm{GT}} \times W_{\mathrm{GT}}$ before scoring. We use bilinear interpolation for RGB frames and nearest-neighbor interpolation for masks. We do not crop, pad, or preserve aspect ratio during metric computation; the same direct-resize rule is applied to every model to avoid model-specific canvas handling.

D.5 Graphic-Subtrack Mask IoU

Perception-Graphic and Formulation-Graphic have objective mask metrics. For P-G, the GT is the object-only white-background artifact. For F-G, the GT is the target-time scene at $t^\star$. In both cases, generated and GT masks are compared after resolution normalization.

For each matched object o, the mask IoU is

$$\mathrm{IoU}_o = \frac{|M_o^{\mathrm{gen}} \cap M_o^{\mathrm{GT}}|}{|M_o^{\mathrm{gen}} \cup M_o^{\mathrm{GT}}| + \epsilon}. \tag{4}$$

For multi-object cases, masks are matched by object identity when available and otherwise by Hungarian matching over center distance and bounding-box overlap. The case-level IoU is the mean over matched objects, with missing objects receiving zero IoU.

真值序列有 N_c^{GT} 解码帧时，最接近物理时刻 $t \in [0, T_c]$ 的帧为

$$i_c(t) = \mathrm{clip}\big(\mathrm{round}(t f_c^{\mathrm{GT}}),\ 0,\ N_c^{\mathrm{GT}} - 1\big)\,. \tag{1}$$

生成视频不将提供方原始时长视为物理时间。提示明确请求 T_c 秒运动，故无论名义为 5 秒、8 秒或其他长度，解码序列均视为 $[0, T_c]$ 上的连续轨迹。若有 N^{gen} 帧，有效评估 fps 为

$$\widehat{f}^{\mathrm{gen}} = \frac{N^{\mathrm{gen}}}{T_c}. \tag{2}$$

最接近物理时刻 t 的生成帧为

$$j_c(t) = \mathrm{clip}\Big(\mathrm{round}(t\widehat{f}^{\mathrm{gen}}),\ 0,\ N^{\mathrm{gen}} - 1\Big)\,. \tag{3}$$

该规则使模型间 fps、时长可比。原始生成 fps 只用于解码；物理时间由提示和案例元数据定义。

示例。真值视频 4 秒、30 fps，共 120 帧；模型返回 8 秒、24 fps，共解码 192 帧。因提示要求"生成 4 秒运动"，将 192 帧解释为 4 秒轨迹，设 $\widehat{f}^{\mathrm{gen}} = 192/4 = 48$ fps。在 $t = 2$ 秒，比较真值帧 $i(2) = 60$ 与生成帧 $j(2) = 96$。物理时间对齐忽略模型原始容器时长 8 秒。

D.3 时间戳网格

各案例存储规范时间戳集合 $\mathcal{T}_c \subset [0, T_c]$。推演指标在相同物理时间比较生成帧与真值帧。统一模型的 $\mathcal{T}_c$ 时间戳即请求关键帧；视频模型密集解码，再对每个 $t \in \mathcal{T}_c$ 采样最近生成帧 $j_c(t)$。碰撞接触、离开斜坡等关键事件时刻存在时，除均匀网格外也加入 $\mathcal{T}_c$。

D.4 分辨率归一化

全部像素、掩码指标在真值分辨率计算。生成图帧尺寸 $H_{\mathrm{gen}} \times W_{\mathrm{gen}}$、真值尺寸 $H_{\mathrm{GT}} \times W_{\mathrm{GT}}$ 时，评分前直接缩放至 $H_{\mathrm{GT}} \times W_{\mathrm{GT}}$。RGB 使用双线性插值，掩码使用最近邻。指标计算不裁剪、补边或保留长宽比；各模型使用相同直接缩放规则，避免模型特定画布处理。

D.5 图形子赛道掩码 IoU

感知–图形与定律表述–图形有客观掩码指标。P-G 真值为白底仅物体图；F-G 真值为 $t^\star$ 的目标场景。均在分辨率归一化后比较生成与真值掩码。

对于各匹配物体 o，掩码 IoU 为

$$\mathrm{IoU}_o = \frac{|M_o^{\mathrm{gen}} \cap M_o^{\mathrm{GT}}|}{|M_o^{\mathrm{gen}} \cup M_o^{\mathrm{GT}}| + \epsilon}. \tag{4}$$

多物体案例优先按身份匹配掩码，否则按中心距离、边界框重叠进行匈牙利匹配。案例 IoU 为匹配物体平均，缺失物体记零。

Table 13: MLLM-judge rubric dimensions. P-G and F-G additionally include objective IoU as described in Section D.5.

Subtrack	Rubric dimensions
P-T	Annotation content, layout, and style/readability
P-G	Background purity, annotation removal, object preservation, and spatial consistency
F-T	Correct option, symbolic formula, numeric substitution, and presentation
F-G	Annotation removal, object match, velocity-arrow quality, and velocity text
Deduction	Annotation removal, object consistency, visual quality, motion smoothness, and physics accuracy

D.6 Invalid Outputs and Rollout Aggregation

Each model is evaluated with three independent rollouts per case and subtrack. Let $s_{m,c,k,r}$ be the score of model m on case c, subtrack k, and rollout r. The rollout-mean score is

$$S_{m,c,k} = \frac{1}{3}\sum_{r=1}^{3} s_{m,c,k,r}. \tag{5}$$

If a rollout produces no decodable artifact, the rollout receives score zero for the affected subtrack. Transient judge or parser failures are retried; persistent failures are recorded in the evaluation log and assigned zero rather than removed from the denominator.

E Metric Definitions and Score Aggregation

E.1 MLLM-Judge Rubrics

All MLLM judge calls use temperature 0. The judge sees the model output, the relevant reference images or metadata, and a subtrack-specific rubric. Criteria are scored on $[0, 1]$. Case-irrelevant criteria are marked as not applicable and excluded from the group mean, with group weights renormalized during score aggregation. Table 13 summarizes the rubric dimensions used by each subtrack. The exact judge prompts, criterion-level fields, and group aggregation rules are provided in Section E.1.1.

For P-G, the spatial-consistency dimension incorporates the programmatic mask IoU from Section D.5. For F-G, the object-match dimension combines the MLLM visual appearance score and the target-state mask IoU:

$$S_{\text{obj}}^{\text{F-G}} = \frac{1}{2} S_{\text{appearance}}^{\text{MLLM}} + \frac{1}{2}\text{IoU}_{\text{F-G}}. \tag{6}$$

This keeps velocity-arrow and velocity-text requirements in the MLLM rubric while grounding object position in an objective mask comparison.

E.1.1 MLLM-Judge Prompt Templates

This section provides the MLLM-judge prompt templates used for subjective scoring. For the four non-Deduction subtracks, unified understanding-generation models and video-generation models are evaluated with the same MLLM-judge prompt templates. The only difference lies in output canonicalization: unified models directly output the final-frame image artifact, whereas video-generation models output an MP4 and the last frame is extracted as the answer artifact.

For Deduction, the two model families use family-specific judge prompts because their outputs have different temporal formats. Video-generation models are judged as full videos using a video-level Deduction prompt. Unified models instead generate timestamped keyframes and are judged frame-by-frame against the corresponding ground-truth frames using a keyframe-level prompt. The keyframe-level prompt shares annotation removal, object consistency, visual quality, and physics accuracy with the video-level prompt, but uses object-position match against the timestamp-aligned ground-truth frame as the keyframe proxy for temporal motion smoothness.

For each evaluated output, the judge is instructed to return only a JSON object. Each scalar field is parsed as a score in the range [0,1], where 0 denotes a completely incorrect or unusable response

表 13：MLLM 量规维度。P-G、F-G 另含 D.5 的客观 IoU。

子赛道	评分维度
P-T	标注内容、布局与样式/可读性
P-G	背景纯度、标注移除、物体保留与空间一致性
F-T	正确选项、符号公式、数值代入与呈现
F-G	标注移除、物体匹配、速度箭头质量与速度文字
推演	标注移除、物体一致性、视觉质量、运动平滑性与物理准确率

D.6 无效输出与多次生成汇总

每模型每案例、每子赛道独立生成三次。设 $s_{m,c,k,r}$ 为模型 m 在案例 c、子赛道 k、第 r 次生成得分。生成平均分为

$$S_{m,c,k} = \frac{1}{3}\sum_{r=1}^{3} s_{m,c,k,r}. \tag{5}$$

生成无可解码结果时，对应子赛道记零。暂时评审、解析失效会重试；持续失效写入日志并记零，不从分母排除。

E 指标定义与评分汇总

E.1 MLLM 评审量规

全部 MLLM 评审调用温度为 0。评审看到模型输出、相关参考图或元数据，以及子赛道专用量规。标准按 $[0, 1]$ 评分；案例无关项标为不适用，从组平均排除，汇总时重新归一化组权重。表 13 总结各子赛道量规维度。精确提示、标准级字段与组汇总规则见 E.1.1 节。

P-G 的空间一致性维度纳入 D.5 的程序掩码 IoU。F-G 的物体匹配维度结合 MLLM 外观评分及目标状态掩码 IoU：

$$S_{\text{obj}}^{\text{F-G}} = \frac{1}{2} S_{\text{appearance}}^{\text{MLLM}} + \frac{1}{2}\text{IoU}_{\text{F-G}}. \tag{6}$$

这保留 MLLM 量规中的速度箭头、文字要求，同时以客观掩码比较锚定位置。

E.1.1 MLLM 评审提示模板

本节提供主观评分的 MLLM 评审提示模板。四个非推演子赛道对统一理解–生成和视频生成模型使用相同模板，区别仅在输出规范化：统一模型直接输出最终帧图像，视频模型输出 MP4，提取末帧作为答案。

推演中两类模型因输出时间格式不同，使用专用评审提示。视频生成模型以视频级提示评审完整视频。统一模型生成带时间戳关键帧，逐帧与对应真值比较，使用关键帧级提示。后者与视频级共享移除标注、物体一致性、视觉质量和物理准确性，但将与时间对齐真值的物体位置匹配作为时间运动平滑性的关键帧代理。

每个输出要求评审仅返回 JSON 对象。各标量字段解析为 [0,1]，范围分数，0 表示完全错误或不可用响应

and 1 denotes a perfect match. A score of -1.0 is reserved for criteria that are not applicable, such as velocity-arrow criteria when no velocity arrow exists in the ground truth. In aggregation, inapplicable criteria are excluded from the corresponding group mean, and the remaining group weights are renormalized. If a response is undecodable or the judge output cannot be parsed after retries, the affected rollout is assigned zero for the corresponding subtrack.

Perception-Text judge prompt. For Perception-Text, the judge compares the generated final-frame annotation artifact against the ground-truth white-background annotation image.

```
You are an expert evaluator assessing how perfectly a generated image preserves the
exact annotation details from a ground truth reference image.

You are provided with two images:
1. "REFERENCE GROUND TRUTH IMAGE": The perfectly correct original image with
annotations on a pure white background.
2. "GENERATED IMAGE": The image to be evaluated.

Your task is to compare the GENERATED IMAGE against the REFERENCE GROUND TRUTH
IMAGE and score the generated image across 18 distinct criteria.
For each criterion, assign a score between 0.0 (completely unmatched/wrong) and 1.0
(perfectly matched).

Evaluate the following criteria carefully:

1. text_font_match: Does every text annotation use the EXACT SAME font family/typeface
as the ground truth?
2. text_style_match: Does every text annotation have the EXACT SAME style
(e.g., bold, italic, regular) as the ground truth?
3. text_size_match: Does every text annotation have the EXACT SAME font size/scale as
the ground truth?
4. text_color_match: Does every text annotation have the EXACT SAME color as the
ground truth?
5. text_position_match: Is every text annotation located in the EXACT SAME position
as the ground truth?
6. text_content_match: Does the textual content of every single annotation EXACTLY
match the ground truth (no typos, missing words, or hallucinations)?
7. text_readability: Are all text labels and annotations perfectly crisp, sharp, and
easy to read, without any blurriness, distortion, or generation artifacts, exactly
matching the clarity of the ground truth? (0.0=heavily blurred/distorted,
1.0=perfectly crisp and readable).
8. axes_color_match: Is the color of the coordinate axes arrows EXACTLY the same as
the ground truth?
9. axes_style_match: Is the line style (solid vs dashed) of the coordinate axes arrows
EXACTLY the same as the ground truth?
10. axes_size_match: Are the width, length, and arrowhead size of the coordinate axes
arrows EXACTLY the same as the ground truth?
11. axes_position_match: Is the position of the coordinate axes EXACTLY the same as
the ground truth?
12. axes_angle_match: Are the angles between the coordinate axes EXACTLY the same as
the ground truth?
13. velocity_color_match: If initial velocity arrows exist in the GT, does the generated
image have velocity arrows with the EXACT SAME color? (If no velocity arrows exist in
GT, score -1.0)
14. velocity_style_match: If initial velocity arrows exist in the GT, do they have the
EXACT SAME line style (solid vs dashed)? (If no velocity arrows exist in GT, score -1.0)
15. velocity_size_match: If initial velocity arrows exist in the GT, do they have the
EXACT SAME width, length, and arrowhead size? (If no velocity arrows exist in GT,
score -1.0)
16. velocity_position_match: If initial velocity arrows exist in the GT, are they in the
EXACT SAME starting position? (If no velocity arrows exist in GT, score -1.0)
17. no_extra_annotations: Does the generated image only contain the annotations present
in the GT, with absolutely NO extra, hallucinated, or duplicated annotations?
(0.0=extra stuff exists, 1.0=clean, matches GT exactly).
18. white_background: Is the background of the entire generated image pure white,
just like the ground truth?

Respond with ONLY JSON:
```


1 表示完美匹配。-1.0 专用于不适用项，如真值不存在速度箭头时的箭头标准。汇总从对应组均值排除不适用项，再归一化剩余组权重。若响应无法解码或重试后评审输出仍无法解析，该次生成对应子赛道记零。

感知–文本评审提示。将生成末帧标注图与真值白底标注图比较。

你是专家评审，判断生成图对真值精确标注细节的保留程度。

给你两图：1.“真值参考图”：纯白背景含标注的完全正确原图；2.“生成图”：待评估图。

任务是将生成图与真值参考比较，按 18 个不同标准评分。各项为 0.0（完全不匹配/错误）至 1.0（完美匹配）。

仔细评估下列标准：

1. text_font_match：每条文本标注的字体家族/字形是否与真值完全相同？2. text_style_match：每条标注样式（粗体、斜体、常规等）是否完全相同？3. text_size_match：每条标注字号/缩放是否完全相同？4. text_color_match：每条标注颜色是否完全相同？5. text_position_match：每条标注位置是否完全相同？6. text_content_match：每条标注文本是否精确匹配真值，无错字、缺词或幻觉？7. text_readability：全部标签是否与真值一样清晰、锐利、易读，无模糊、变形或生成伪影？（0.0=：严重模糊/变形；1.0=：完全清晰可读。）8. axes_color_match：坐标轴箭头颜色是否完全相同？9. axes_style_match：坐标轴箭头线型（实线/虚线）是否完全相同？10. axes_size_match：坐标轴箭头宽度、长度、箭头头部大小是否完全相同？11. axes_position_match：坐标轴位置是否完全相同？12. axes_angle_match：坐标轴夹角是否完全相同？13. velocity_color_match：若真值有初速度箭头，生成图是否有完全相同颜色的箭头？无箭头则记 -1.0。14. velocity_style_match：若真值有初速度箭头，线型是否完全相同？无箭头则 -1.0。15. velocity_size_match：若真值有初速度箭头，宽度、长度及头部大小是否完全相同？无箭头则 -1.0。16. velocity_position_match：若真值有初速度箭头，起点位置是否完全相同？无箭头则 -1.0。17. no_extra_annotations：生成图是否仅含真值标注，完全没有额外、幻觉或重复标注？（0.0=：存在额外内容；1.0=：干净且精确匹配。）18. white_background：整幅生成图背景是否与真值一样纯白？

仅以 JSON 格式回答：

```
{
  "text_font_match": <float>,
  "text_style_match": <float>,
  "text_size_match": <float>,
  "text_color_match": <float>,
  "text_position_match": <float>,
  "text_content_match": <float>,
  "text_readability": <float>,
  "axes_color_match": <float>,
  "axes_style_match": <float>,
  "axes_size_match": <float>,
  "axes_position_match": <float>,
  "axes_angle_match": <float>,
  "velocity_color_match": <float>,
  "velocity_style_match": <float>,
  "velocity_size_match": <float>,
  "velocity_position_match": <float>,
  "no_extra_annotations": <float>,
  "white_background": <float>,
  "feedback": "<2-4 sentences summarizing any deviations found across these criteria>"
}
```

The Perception-Text score is computed from three groups: content, layout, and style. The content group averages text content match, text readability, and no-extra-annotation score. The layout group averages text position, axes position, axes angle, and velocity position. The style group averages text font, text style, text size, text color, axes color, axes style, axes size, velocity color, velocity style, velocity size, and white background. The final Perception-Text score is a weighted average of these three groups with weights 0.50, 0.30, and 0.20.

Perception-Graphic judge prompt. For Perception-Graphic, the judge receives the annotation-only reference, the object-only reference, and the generated object-only artifact.

```
You are an expert image evaluator.

You are provided with THREE images:
1. "REFERENCE WHITE BACKGROUND ANNOTATION IMAGE": a ground-truth reference on a
pure white background that contains human annotations (coordinate axes, text labels,
and possibly initial-velocity arrows).
2. "REFERENCE WHITE BACKGROUND OBJECT IMAGE": a ground-truth reference on a pure
white background that contains ONLY the physical objects (no annotations, no
background/ground/environment).
3. "GENERATED IMAGE": the image to be evaluated.

Your job is to score how well the GENERATED IMAGE matches the intended "objects-only
on pure white" result, using the two reference images to understand what should be
removed and what should remain.

Score each metric from 0.0 to 1.0 (higher is better). Be strict: partial visibility,
faint remnants, blur, or mismatched appearance should reduce the score.

BACKGROUND (purity):
- white_background: Is the entire background in the GENERATED IMAGE truly pure white
(no gradients, shadows, textures, ground plane, or environmental remnants)?

ANNOTATION REMOVAL (compare against the annotation reference):
- coord_axes: Are coordinate axes and axis labels fully removed?
(0=clearly visible, 1=completely gone)
- velocity_arrows: Are any initial-velocity arrows fully removed?
(0=visible, 1=gone. If NO velocity arrows exist in the annotation reference,
return -1.0.)
- text_annotations: Are ALL text labels, numbers, and other textual annotations removed?
(0=visible, 1=gone)

OBJECT PRESERVATION (compare against the object reference):
- objects_only_from_gt: Does the GENERATED IMAGE contain ONLY the physical objects
present in the object reference (no extra objects, no missing objects, and no leftover
ground/background surfaces)?
```


{ "text_font_match": <float>, "text_style_match": <float>, "text_size_match": <float>, "text_color_match": <float>, "text_position_match": <float>, "text_content_match": <float>, "text_readability": <float>, "axes_color_match": <float>, "axes_style_match": <float>, "axes_size_match": <float>, "axes_position_match": <float>, "axes_angle_match": <float>, "velocity_color_match": <float>, "velocity_style_match": <float>, "velocity_size_match": <float>, "velocity_position_match": <float>, "no_extra_annotations": <float>, "white_background": <float>, "feedback": "<2-4 sentences summarizingany deviations found across these criteria>" }

感知–文本评分分为内容、布局、样式三组。内容组平均文本内容匹配、可读性、无额外标注；布局组平均文本位置、坐标轴位置、轴夹角、速度箭头位置；样式组平均字体、文字样式、字号、文字颜色、轴颜色/样式/大小、速度箭头颜色/样式/大小、白背景。最终三组加权平均，权重为 0.50、0.30、0.20。

感知—图形评审提示。评审接收标注参考、仅物体参考及生成仅物体图。

```
你是专家图像评审。

给你三张图：1.“白底标注参考图”：纯白背景真值参考，含人工标注（坐标轴、文本标签，可能有
初速度箭头）。2.“白底物体参考图”：纯白背景真值参考，只含物体，无标注、背景/地面/环境。
3.“生成图”：待评估图像。

任务是评估生成图与预期“纯白底仅物体”结果的匹配程度，利用两参考图判断应移除和保留内容。

各指标评分 0.0–1.0，越高越好。严格评分：部分可见、淡残影、模糊、外观不匹配应扣分。

背景纯度：- white_background：整个背景是否真正纯白，无渐变、阴影、纹理、地面或环境残留?

标注移除（对比标注参考）：- coord_axes：坐标轴及轴标签是否完全移除？（0=：明显可见； 1=：完全
消失。）- velocity_arrows：初速度箭头是否完全移除？（0=：可见； 1=：消失。参考无速度箭头则返
回 -1.0。）- text_annotations：全部文字、数字及其他文字标注是否移除？（0=：可见； 1=：消失。）

物体保留（与物体参考比较）：
- objects_only_from_gt：生成图是否只含物体参考中的实体，无额外、缺失物体、残留地面/背景?
```

```
- object_completeness: Are all objects from the object reference present and
uncut/undamaged in the GENERATED IMAGE (no missing parts, no erasures, no merging into
the background)?
- object_visual_match: Do the kept objects match the object reference visually
(shape/geometry, relative size, colors, textures/materials, shading/lighting cues, and
any distinctive visual details)? (0=looks different, 1=visually identical)
- object_position_match: Do the objects appear at the EXACT SAME image positions as in
the object reference (same 2D placement and relative arrangement; no shifting, drifting,
or rearrangement)?

Respond with ONLY JSON:
{
  "white_background": <float>,
  "coord_axes": <float>,
  "velocity_arrows": <float>,
  "text_annotations": <float>,
  "objects_only_from_gt": <float>,
  "object_completeness": <float>,
  "object_visual_match": <float>,
  "object_position_match": <float>,
  "feedback": "<2-4 sentences summarizing the biggest issues>"
}
```

The Perception-Graphic score is computed from four groups: background purity, annotation removal, object preservation, and spatial consistency. The background group contains white background. The annotation-removal group averages coordinate axes, velocity arrows, and text annotations. The object-preservation group averages objects-only-from-ground truth, object completeness, and object visual match. The spatial group averages object position match and the programmatic segmentation IoU. The final Perception-Graphic score uses group weights 0.10, 0.30, 0.30, and 0.30.

Formulation-Text judge prompt. For Formulation-Text, the judge evaluates whether the final white-background answer contains the correct option, the correct symbolic law, and a correct numeric substitution.

```
You are evaluating a model's physical comprehension.

The model was asked to output an image containing 3 text lines on a pure white background:
Line 1: The correct option letter.
Line 2: The correct formula.
Line 3: The formula with annotated variables replaced by corresponding values. If the
scene has multiple objects, values belonging to specific objects must come from exactly
ONE consistent set (e.g., one object for single-body motion, or multiple objects forming
one valid set for multiple-body motion).

Correct Formula is {correct_letter}: {correct_formula}
Annotation Content: {annotation}

Evaluate the generated image (Score 0.0-1.0):

- option_correct: Is the correct letter (Line 1) chosen?
- formula_variables_correct: Are physical symbols in the formula (Line 2) written
correctly?
- formula_constants_correct: Are inherent numeric constants in the formula written
correctly?
- formula_operators_correct: Are operators in the formula (Line 2) written correctly?
- substitution_all_replaced: Are ALL annotated variables replaced by values in Line 3?
- substitution_values_correct: Are substituted values correct? If there are multiple
objects, did the model pick exactly one valid set of objects to substitute values?
- substitution_unreplaced_correct: Are unannotated symbols correctly kept unchanged in
Line 3?
- substitution_constants_correct: Are inherent numeric constants correctly kept in
Line 3?
- substitution_operators_correct: Are operators correctly written in Line 3?
- format_3_lines: Does it strictly follow the 3-line format?
- background_pure_white: Is the background pure white?

Respond with ONLY JSON:
```


- object_completeness：参考全部物体是否在生成图存在且未裁切/损坏，无缺部件、擦除或融入背景?

- object_visual_match：保留物体是否在视觉上匹配参考，包括形状/几何、相对大小、颜色、纹理/材料、阴影/照明及独特细节？（0=：不同； 1=：视觉完全相同。）- object_position_match：物体是否位于参考中完全相同的图像位置，即二维放置和相对排列相同，无偏移、漂移

，

或重排）？

Respond with ONLYJSON: { "white_background": <float>, "coord_axes": <float>, "velocity_arrows": <float>, "text_annotations": <float>, "objects_only_from_gt": <float>, "object_completeness": <float>, "object_visual_match": <float>, "object_position_match": <float>, "feedback": "<2-4 sentences summarizingthe biggest issues>" }

感知–图形评分分为背景纯度、标注移除、物体保留、空间一致性四组。背景组含白背景；标注移除组平均坐标轴、速度箭头、文字标注；物体保留组平均只含真值物体、物体完整性、物体视觉匹配；空间组平均物体位置匹配和程序分割 IoU。最终权重为 0.10、0.30、0.30、0.30。

定律表述–文本评审提示。评估最终白底答案是否含正确选项、正确符号定律、正确数值代入。

你正在评估模型物理理解能力。

要求模型输出纯白背景含三行文字的图：第 1 行，正确选项字母。

第 2 行：正确公式。
第 3 行：将公式标注变量替换为对应值。多物体场景中，属于特定物体的值须来自恰好一组一致对象，如单体运动选一个物体，多体运动选组成一组有效组合的多个物体。

正确公式为 {correct_letter}: {correct_formula}；标注内容：
{annotation}

评估生成图（0.0–1.0）：

- option_correct：第 1 行是否选对字母？- formula_variables_correct：第 2 行公式的物理符号是否正确？- formula_constants_correct：固有数值常量是否正确？- formula_operators_correct：第 2 行运算符是否正确？- substitution_all_replaced：第 3 行是否替换全部标注变量为数值？- substitution_values_correct：代入数值是否正确？多物体时是否恰选一组有效物体代入？- substitution_unreplaced_correct：第 3 行未标注符号是否正确保持不变？- substitution_constants_correct：第 3 行固有数值常量是否正确保留？- substitution_operators_correct：第 3 行运算符是否正确？- format_3_lines：是否严格遵循三行格式？- background_pure_white：背景是否纯白?

仅以 JSON 格式回答：

```
{
  "option_correct": <float>,
  "formula_variables_correct": <float>,
  "formula_constants_correct": <float>,
  "formula_operators_correct": <float>,
  "substitution_all_replaced": <float>,
  "substitution_values_correct": <float>,
  "substitution_unreplaced_correct": <float>,
  "substitution_constants_correct": <float>,
  "substitution_operators_correct": <float>,
  "format_3_lines": <float>,
  "background_pure_white": <float>,
  "feedback": "<2-3 sentences summarizing issues>"
}
```

The Formulation-Text score is computed from four groups: correct option, symbolic formula, numeric substitution, and presentation. The symbolic-formula group averages formula variables, constants, and operators. The numeric-substitution group averages whether all annotated variables are replaced, whether substituted values are correct, whether unannotated symbols are preserved, whether constants are preserved, and whether operators are correct. The presentation group averages the three-line format score and the pure-white-background score. The final Formulation-Text score uses group weights 0.20, 0.30, 0.40, and 0.10.

Formulation-Graphic judge prompt. For Formulation-Graphic, the judge compares the generated target-time scene against the ground-truth target-time reference and the initial annotated scene.

```
You are evaluating a generated image against a ground truth (GT) reference image and an
initial annotated scene image.

The images are provided in this order:
1. REFERENCE GROUND TRUTH IMAGE (showing the correct physical positions of all objects,
their velocity arrow directions, and their instantaneous speed values).
2. INITIAL ANNOTATED PHYSICS SCENE (showing objects, axes, and all initial text labels).
3. GENERATED IMAGE.

Please evaluate the GENERATED IMAGE based on the following metrics. Score each from
0.0 to 1.0. If there are multiple objects, your score should reflect the average
performance across ALL objects:

- annotations_removed: Are all annotations from the INITIAL ANNOTATED PHYSICS SCENE
(coordinate axes, text labels, initial velocity arrows) correctly removed in the
GENERATED IMAGE? (0=not removed at all, 1=completely clean)
- object_position_match: Are the positions of all objects in the GENERATED IMAGE exactly
the same as their corresponding positions in the GT reference image?
(0=completely different, 1=exact match for all objects)
- object_appearance_match: Do all objects in the GENERATED IMAGE look visually identical
to the corresponding objects in the GT reference image, with special attention to EXACT
texture/material detail and EXACT object size/scale? (0=very different, 1=identical for
all objects)
- arrow_is_borderless_orange: Are the instantaneous velocity arrows for all moving
objects in the GENERATED IMAGE borderless orange arrows? (0=no/wrong style/color for all,
1=yes for all)
- arrow_from_center: Do the instantaneous velocity arrows in the GENERATED IMAGE
originate from the centers of their respective objects? (0=no for all, 1=yes for all)
- arrow_direction_match: Are the directions of the instantaneous velocity arrows in the
GENERATED IMAGE completely consistent with the corresponding velocity arrow directions
in the GT reference image? (0=completely wrong directions, 1=exact match for all moving
objects)
- velocity_label_present_and_format: Does EVERY object in the GENERATED IMAGE have a
velocity text label that roughly follows the pattern "v = number" (or "v = 0" if at rest)?
The exact spacing/decimals do NOT need to match; the key is that a clear velocity label
exists. (0=missing labels for all objects, 1=labels present for all objects)
- velocity_value_match: How closely do the numerical values of the velocity text labels
in the GENERATED IMAGE match the values in the GT reference image? (0=values are
completely wrong or missing, 1=values match exactly for all objects. Give partial credit,
e.g., 0.1-0.9, depending on how numerically close the values are and how many objects are
correct)
```



```
{ "option_correct": <float>, "formula_variables_
correct": <float>, "formula_constants_correct": <
float>, "formula_operators_correct": <float>, "
substitution_all_replaced": <float>, "substitution_
values_correct": <float>, "substitution_unreplaced_
correct": <float>, "substitution_constants_correct": <
float>, "substitution_operators_correct": <float>, "
format_3_lines": <float>, "background_pure_white": <
float>, "feedback": "<2-3 sentences
summarizingissues>" }
```

定律表述–文本评分分为正确选项、符号公式、数值代入、呈现四组。符号公式组平均变量、常量、运算符；数值代入组平均是否替换全部标注变量、代入值是否正确、未标注符号是否保留、常量是否保留、运算符是否正确；呈现组平均三行格式和纯白背景。最终组权重为 0.20、0.30、0.40、0.10。

定律表述–图形评审提示。将生成目标场景与目标时刻真值、初始标注场景比较。

```
你将生成图与真值参考及初始标注场景比较。

图像按以下顺序提供：
1. 真值参考图（展示全部物体正确位置、速度箭头方向和瞬时速率）；2. 初始标注物理场景（物体、坐标
轴及全部初始文本）；3. 生成图。

按下列指标评生成图，各项 0.0–1.0。多物体时评分反映全部物体平均表现：

— 标注 _移除：初始标注物理场景的全部标注是否
（坐标轴、文本、初速度箭头）是否正确移除于
生成图中？（0=：完全未移除； 1=：完全干净。）
- object_position_match：生成图全部物体位置是否精确
与真值参考对应位置相同?
（0=：完全不同； 1=：全部精确匹配。）
- object_appearance_match：生成图全部物体是否与真值对应物体视觉完全相同，尤其精确纹理/材质细
节和精确大小/缩放？（0=：非常不同； 1=：全部相同。）

- arrow_is_borderless_orange：全部运动物体瞬时速度箭头是否无边框橙色？（0=：全部否/样式颜色错
误；1=：全部是。）

- arrow_from_center：生成图瞬时速度箭头是否从各物体中心出发？（0=：全部否； 1=：全部是。）-
arrow_direction_match：箭头方向是否与真值对应速度箭头完全一致？（0=：完全错误； 1=：全部运
动物体精确匹配。）

- velocity_label_present_and_format：生成图每个物体是否有大致遵循“v = 数字”（静止则“v = 0”）
的速度标签？空格、小数位无需完全相同，关键是清晰标签存在。（0=：全部缺失； 1=：全部存在。）

- velocity_value_match：生成图速度标签数值与真值多接近？（0=：完全错误或缺失； 1=：全部精确匹
配。按数值接近程度及正确物体数给 0.1–0.9 等部分分。）
```

```
- velocity_label_no_unit: Do the velocity text labels in the GENERATED IMAGE contain only
numbers WITHOUT any units (e.g., no "m/s" or similar)? (0=all have units, 1=no units for
any object)

Respond with ONLY JSON:
{
  "annotations_removed": <float>,
  "object_position_match": <float>,
  "object_appearance_match": <float>,
  "arrow_is_borderless_orange": <float>,
  "arrow_from_center": <float>,
  "arrow_direction_match": <float>,
  "velocity_label_present_and_format": <float>,
  "velocity_value_match": <float>,
  "velocity_label_no_unit": <float>,
  "feedback": "<2-4 sentences summarizing issues>"
}
```

The Formulation-Graphic score is computed from four groups: annotation removal, object match, velocity-arrow quality, and velocity text. The object-match group averages object position match, object appearance match, and the target-state segmentation IoU. The velocity-arrow group averages arrow style/color, arrow origin, and arrow direction. The velocity-text group averages velocity-label presence and format, velocity-value match, and absence of units. The final Formulation-Graphic score uses group weights 0.10, 0.30, 0.30, and 0.30.

Deduction judge prompt for video-generation models. For video-generation models, the judge evaluates the full generated video. When ground-truth frames are available to the judge, the optional ground-truth comparison section is inserted through the placeholder.

```
You are evaluating a physics video generation model's ability to generate physically
accurate and visually high-quality motion.

You are provided with:
1. INITIAL ANNOTATED PHYSICS SCENE (the first frame, showing objects, coordinate axes,
and annotated initial conditions).
2. GENERATED VIDEO (the model's output).
{gt_section}

Please evaluate the GENERATED VIDEO based on the following metrics. Score each from
0.0 to 1.0:

- gen_annotations_removed: Are all human annotations from the INITIAL SCENE (such as
text labels, coordinate axes, and initial velocity arrows) successfully removed in the
generated video? (0 = annotations remain visible or leave severe ghostly artifacts,
1 = perfectly clean removal, leaving only the physical objects and background).
- object_consistency: Do the physical objects remain perfectly consistent throughout the
video? (0 = objects suddenly disappear, new objects appear out of nowhere, or objects
undergo severe unintended deformation, 1 = objects maintain their exact identity, shape,
and count perfectly).
- visual_quality: What is the overall visual quality of the generated video? Consider
sharpness, clarity, and absence of visual artifacts or noise. (0 = severe artifacts or
blur, 1 = high-quality, crisp rendering).
- motion_smoothness: Is the temporal motion of the objects smooth and natural?
(0 = severely broken motion, flickering, stuttering, or teleporting, 1 = perfectly
smooth, continuous, and natural movement).
- physics_accuracy: Does the motion of the objects accurately follow the physics
described by the initial conditions (trajectory, velocity, gravity, collisions, etc.)?
(0 = completely defies physics, 1 = physically accurate motion).

Respond with ONLY JSON:
{
  "gen_annotations_removed": <float>,
  "object_consistency": <float>,
  "visual_quality": <float>,
  "motion_smoothness": <float>,
  "physics_accuracy": <float>,
  "feedback": "<2-4 sentences summarizing issues regarding annotation removal, visual
```


- velocity_label_no_unit：生成图速度标签是否只含数值，无 m/s 等单位？（0=：全部有单位； 1=：全部无单位。）

Respond with ONLYJSON: { "annotations_removed": <float>, "object_position_match": <float>, "object_appearance_match": <float>, "arrow_is_borderless_orange": <float>, "arrow_from_center": <float>, "arrow_direction_match": <float>, "velocity_label_present_and_format": <float>, "velocity_value_match": <float>, "velocity_label_no_unit": <float>, "feedback": "<2-4 sentences summarizingissues>" }

定律表述–图形评分分为标注移除、物体匹配、速度箭头质量、速度文字四组。物体匹配组平均位置匹配、外观匹配、目标状态分割 IoU；速度箭头组平均样式/颜色、起点、方向；速度文字组平均标签存在与格式、速度数值匹配、无单位。最终权重为 0.10、0.30、0.30、0.30。

视频模型的推演评审提示。评审完整视频；真值帧可用时，通过占位符插入可选真值比较部分。

你评估物理视频生成模型产生物理准确、高视觉质量运动的能力。

提供给你的内容包括：

1. 初始标注物理场景（首帧含物体、坐标轴、标注初始条件）；2. 生成视频（模型输出）。{gt_section}

按下列指标评估生成视频，各项 0.0–1.0：

- gen_annotations_removed：视频是否成功移除初始场景全部人工标注，如文字、坐标轴、初速度箭头？（0 =：仍可见或残留严重幽灵伪影；1 = ：完全干净移除，仅余物体和背景。）

- object_consistency：物体是否在全视频保持完全一致？（0 = ：突然消失、凭空出现或严重非预期变形； 1 =：身份、形状、数量完全保持。）

- visual_quality：整体视频质量如何？考虑锐度、清晰度、无伪影或噪声。（0 = ：严重伪影模糊；1 = ：高质量清晰渲染。）

- motion_smoothness：物体时间运动是否平滑自然？（0 = ：严重断裂、闪烁、卡顿、瞬移；1 = ：完全平滑连续自然。）

- physics_accuracy：运动是否准确遵循初始条件规定的轨迹、速度、重力、碰撞等物理规律？（0 = ：完全违反物理； 1 = ：准确物理运动。）

Respond with ONLYJSON: { "gen_annotations_removed": <float>, "object_consistency": <float>, "visual_quality": <float>, "motion_smoothness": <float>, "physics_accuracy": <float>, "feedback": "<2-4 sentences summarizingissues regarding annotation removal, visual

```
  quality, object consistency, smoothness, and physics.>"
}
```

The optional ground-truth comparison section is:

```
3. GROUND TRUTH FRAMES showing the correct physics are also provided. Compare generated
vs GT: object positions, trajectories, timing.
```

Deduction keyframe judge prompt for unified understanding-generation models. For unified understanding-generation models that output timestamped keyframes rather than dense videos, the judge evaluates each generated keyframe against the corresponding ground-truth frame.

```
You are evaluating a single keyframe from a physics simulation generation model.

You are provided with:
1. INITIAL ANNOTATED PHYSICS SCENE (the first frame with coordinate axes, text labels,
etc.)
2. GROUND TRUTH FRAME at t = {time_point:.2f}s (the correct physical state at this
moment)
3. GENERATED FRAME at t = {time_point:.2f}s (the model's output for this moment)

Compare the GENERATED FRAME against the GROUND TRUTH FRAME. Score each from 0.0 to 1.0:

- gen_annotations_removed: Are all human annotations from the INITIAL SCENE (text labels,
coordinate axes, initial velocity arrows) removed in the generated frame?
(0 = annotations clearly visible, 1 = perfectly clean)
- object_consistency: Do the physical objects match the GT in identity, count, and shape?
(0 = objects missing/hallucinated/severely deformed, 1 = exact match)
- visual_quality: Overall visual quality -- sharpness, clarity, absence of artifacts.
(0 = severe artifacts/blur, 1 = high quality crisp rendering)
- object_position_match: Are the objects at the same positions as in the GT frame?
(0 = completely wrong positions, 1 = exact position match)
- physics_accuracy: Does the overall physical state (positions, deformations,
interactions) match the GT? (0 = completely wrong, 1 = physically accurate)

Respond with ONLY JSON:
{
  "gen_annotations_removed": <float>,
  "object_consistency": <float>,
  "visual_quality": <float>,
  "object_position_match": <float>,
  "physics_accuracy": <float>,
  "feedback": "<1-2 sentences on the biggest issues>"
}
```

For Deduction, the subjective MLLM scores are combined with programmatic objective metrics. For video-generation models, annotation removal and object consistency form the integrity group; visual quality, motion smoothness, normalized PSNR, and masked PSNR form the fidelity group; physics accuracy, Spatial IoU, Spatiotemporal IoU, Weighted Spatial IoU, and velocity accuracy form the physics group. The final Deduction score uses group weights 0.20 for integrity, 0.20 for fidelity, and 0.60 for physics.

For unified understanding-generation models, the keyframe-level scores are first averaged across the requested timestamps. The object-position-match field is then used as the keyframe-level proxy for the shared motion-smoothness field before applying the same Deduction score aggregation.

E.2 Deduction Objective Metrics

Deduction is evaluated at the timestamp set $\mathcal{T}_c$. For every timestamp, the generated frame is selected by the time-rescaling rule in Section D.2, resized to GT resolution, and compared against the corresponding GT frame.

Normalized PSNR. We compute frame-level PSNR between generated and GT RGB frames and normalize by clipping at 40 dB:

$$S_{\text{PSNR}} = \min(\text{PSNR}/40, 1). \tag{7}$$


```
  质量、物体一致性、平滑性与物理。>”
}
```

可选真值比较部分为：

```
3. 另提供展示正确物理的真值帧。比较生成与真值的位置、轨迹、时序。
```

统一理解–生成模型的推演关键帧评审提示。对于输出时间戳关键帧而非密集视频的模型，逐关键帧与对应真值评估。

```
你评估物理仿真生成模型的单个关键帧。

输入为：1. 初始标注物理场景（含坐标轴、文本等的首帧）；2. t = {time_point:.2f} 秒的真值帧（该
时刻正确物理状态）；3. t ={time_point:.2f} 秒的生成帧（模型输出）。

比较生成帧与真值帧，各项评分 0.0–1.0:

- gen_annotations_removed：生成帧是否移除初始场景全部人工标注（文本、坐标轴、初速度箭头）？
（0 = ：标注明显可见； 1 = ：完全干净。）- object_consistency：物体身份、数量和形状是否与真值
一致？（0 = ：缺失/幻觉/严重变形； 1 = ：精确匹配。）- visual_quality：整体视觉质量，包括锐度、
清晰度及无伪影。（0 = ：严重伪影/模糊； 1 =：高质量清晰渲染。）- object_position_match：物体位
置是否与真值帧相同？（0 = ：完全错误； 1 = ：精确匹配。）- physics_accuracy：整体物理状态（位
置、变形、交互）是否匹配真值？（0 = ：完全错误； 1 = ：物理准确。）

Respond with ONLYJSON: { "gen_annotations_removed":
 <float>, "object_consistency": <float>, "visual_quality": <
float>, "object_position_match": <float>, "physics_
accuracy": <float>, "feedback": "<1-2 sentences on
thebiggest issues>" }
```

推演将主观 MLLM 评分与程序客观指标结合。视频模型中，标注移除和物体一致性构成完整性组；视觉质量、运动平滑性、归一化 PSNR、掩码 PSNR 构成保真组；物理准确性、空间 IoU、时空 IoU、加权空间 IoU、速度准确性构成物理组。最终权重为完整性 0.20、保真 0.20、物理 0.60。

统一理解–生成模型先对请求时间戳平均关键帧评分，再用物体位置匹配作为共同运动平滑性字段的关键帧代理，应用同一推演汇总。

E.2 推演客观指标

推演在时间戳集合 $\mathcal{T}_c$ 评估。每时刻按 D.2 时间缩放规则选生成帧，缩至真值分辨率，对对应真值帧比较。

归一化 PSNR。逐帧计算生成与真值 RGB 的 PSNR，以 40 dB 截断归一化：

$$S_{\text{PSNR}} = \min(\text{PSNR}/40, 1). \tag{7}$$

Masked PSNR. Masked PSNR is computed only on the union of GT foreground object masks. This reduces background dominance and focuses the metric on object placement and appearance.

Motion-mask extraction. Following Physics-IQ [41], the three IoU-based Deduction metrics are computed on *motion/action masks*, not on raw RGB frames. After applying the time-rescaling rule of Section D.2 and resizing generated frames to the GT resolution, we use SAM3 to segment the physically relevant moving objects in both the generated clip and the GT clip, yielding binary mask sequences

$$\mathcal{A}^{\mathrm{gen}} = \{A_t^{\mathrm{gen}}\}_{t\in\mathcal{T}_c}, \qquad \mathcal{A}^{\mathrm{GT}} = \{A_t^{\mathrm{GT}}\}_{t\in\mathcal{T}_c}.$$

Here $A_t(p) \in \{0,1\}$ indicates whether pixel p belongs to a physically relevant moving object at physical time t. For simulator cases, the engine-provided instance masks are used as GT masks when available; for real-world cases, the human-refined SAM3 masks are used as GT masks. For generated videos, SAM3 is applied with the same object prompts and post-processing rules as the GT annotation pipeline. The resulting masks are therefore object/action masks: they measure the spatial support of the law-relevant moving bodies, rather than exact RGB texture or background appearance.

Spatial IoU. Spatial IoU answers the Physics-IQ question: *where does action happen?* For each clip, we collapse the binary motion-mask sequence along the time dimension with a max operation:

$$S^x(p) = \max_{t\in\mathcal{T}_c} A_t^x(p), \qquad x \in \{\mathrm{gen}, \mathrm{GT}\}.$$

Thus $S^x(p) = 1$ if any motion occurs at pixel p at any evaluated time, and 0 otherwise. The Spatial IoU is then

$$S_{\mathrm{spatialIoU}} = \frac{\sum_p \mathbf{1}\big[S^{\mathrm{gen}}(p) = 1 \wedge S^{\mathrm{GT}}(p) = 1\big]}{\sum_p \mathbf{1}[S^{\mathrm{gen}}(p) = 1 \vee S^{\mathrm{GT}}(p) = 1] + \epsilon}.$$

This metric ignores the exact timing of the action and evaluates whether the generated video places motion in the same spatial regions as the GT video.

Spatiotemporal IoU. Spatiotemporal IoU answers the Physics-IQ question: *where and when does action happen?* Instead of collapsing time, we compare the generated and GT motion masks at each aligned physical timestamp and average the resulting frame-wise IoUs:

$$S_{\mathrm{stIoU}} = \frac{1}{|\mathcal{T}_c|} \sum_{t\in\mathcal{T}_c} \frac{\sum_p \mathbf{1}\big[A_t^{\mathrm{gen}}(p) = 1 \wedge A_t^{\mathrm{GT}}(p) = 1\big]}{\sum_p \mathbf{1}\big[A_t^{\mathrm{gen}}(p) = 1 \vee A_t^{\mathrm{GT}}(p) = 1\big] + \epsilon}.$$

A model can therefore obtain high Spatial IoU but low Spatiotemporal IoU if it produces action in the right places but at the wrong time.

Weighted Spatial IoU. Weighted Spatial IoU answers the Physics-IQ question: *where and how much action happens?* Rather than using a binary "ever-moved" spatial map, we collapse each motion-mask sequence by temporal averaging:

$$W^x(p) = \frac{1}{|\mathcal{T}_c|} \sum_{t\in\mathcal{T}_c} A_t^x(p), \qquad x \in \{\mathrm{gen}, \mathrm{GT}\}.$$

Here $W^x(p) \in [0,1]$ measures how frequently pixel p is active over the evaluated clip. The weighted spatial overlap is computed by the generalized IoU ratio used in Physics-IQ:

$$S_{\mathrm{wIoU}} = \frac{\sum_p \min\big(W^{\mathrm{gen}}(p), W^{\mathrm{GT}}(p)\big)}{\sum_p \max(W^{\mathrm{gen}}(p), W^{\mathrm{GT}}(p)) + \epsilon}.$$

Unlike Spatial IoU, this metric distinguishes transient motion from repeated or prolonged motion at the same location. For example, a ball passing through a region once and an object repeatedly oscillating through that region may share a similar binary spatial support, but they induce different weighted motion maps.

Velocity accuracy. For sampled frames where 3D velocity can be estimated, we recover object-level velocity vectors and compare them with the law-predicted velocity. If the mean Euclidean velocity error is e_v, the velocity score is

$$S_{\mathrm{vel}} = \frac{1}{1+e_v}. \tag{8}$$

This maps larger errors smoothly toward zero while preserving a $[0,1]$ score range.

掩码 PSNR。只在真值前景物体掩码并集计算，减少背景主导，聚焦物体位置与外观。

运动掩码提取。遵循 Physics-IQ [41], ，三种基于 IoU 的推演指标在运动/动作掩码而非原始 RGB 上计算。应用 D.2 时间缩放规则并将生成帧缩放至真值分辨率后，对生成与真值片段均用 SAM3 分割物理相关运动物体，得到二值掩码序列

$$\mathcal{A}^{\mathrm{gen}} = \{A_t^{\mathrm{gen}}\}_{t\in\mathcal{T}_c}, \qquad \mathcal{A}^{\mathrm{GT}} = \{A_t^{\mathrm{GT}}\}_{t\in\mathcal{T}_c}.$$

这里 $A_t(p) \in \{0,1\}$ 表示像素 p 在物理时间 t 是否属于物理相关运动物体。仿真案例在可用时采用引擎实例掩码作为真值；真实案例采用人工修正的 SAM3 掩码。生成视频也使用相同物体提示、后处理规则运行 SAM3。因此掩码是物体/动作掩码，测量定律相关运动体的空间区域，而非精确 RGB 纹理或背景外观。

空间 IoU。回答 Physics-IQ 的“动作在哪里发生？”问题。每片段以最大操作沿时间折叠二值运动掩码：

$$S^x(p) = \max_{t\in\mathcal{T}_c} A_t^x(p), \qquad x \in \{\mathrm{gen}, \mathrm{GT}\}.$$

任一评估时刻像素 p 有运动时， $S^x(p) = 1$；否则为 0。空间 IoU 为

$$S_{\mathrm{spatialIoU}} = \frac{\sum_p \mathbf{1}\big[S^{\mathrm{gen}}(p) = 1 \wedge S^{\mathrm{GT}}(p) = 1\big]}{\sum_p \mathbf{1}[S^{\mathrm{gen}}(p) = 1 \vee S^{\mathrm{GT}}(p) = 1] + \epsilon}.$$

该指标忽略精确动作时序，评估生成运动是否出现在真值相同空间区域。

时空 IoU。回答 Physics-IQ 的“动作在何处、何时发生？”问题。不折叠时间，而在每个对齐物理时间戳比较生成与真值运动掩码，平均逐帧 IoU：

$$S_{\mathrm{stIoU}} = \frac{1}{|\mathcal{T}_c|} \sum_{t\in\mathcal{T}_c} \frac{\sum_p \mathbf{1}\big[A_t^{\mathrm{gen}}(p) = 1 \wedge A_t^{\mathrm{GT}}(p) = 1\big]}{\sum_p \mathbf{1}\big[A_t^{\mathrm{gen}}(p) = 1 \vee A_t^{\mathrm{GT}}(p) = 1\big] + \epsilon}.$$

因此，正确位置、错误时间的动作可得高空间 IoU，却有低时空 IoU。

加权空间 IoU。回答 Physics-IQ 的“动作在何处、发生多少？”问题。不使用二值“曾运动”图，而沿时间平均各运动掩码序列：

$$W^x(p) = \frac{1}{|\mathcal{T}_c|} \sum_{t\in\mathcal{T}_c} A_t^x(p), \qquad x \in \{\mathrm{gen}, \mathrm{GT}\}.$$

这里 $W^x(p) \in [0,1]$ 衡量像素 p 在片段中活跃频率。加权空间重叠用 Physics-IQ 的广义 IoU 比值：

$$S_{\mathrm{wIoU}} = \frac{\sum_p \min\big(W^{\mathrm{gen}}(p), W^{\mathrm{GT}}(p)\big)}{\sum_p \max(W^{\mathrm{gen}}(p), W^{\mathrm{GT}}(p)) + \epsilon}.$$

不同于空间 IoU，该指标区分同位置的瞬时运动与重复/持续运动。例如球只通过区域一次与物体在区域反复振荡，可能有相似二值空间区域，却诱导不同加权运动图。

速度准确性。在可估计三维速度的采样帧，恢复物体级速度向量，对定律预测速度比较。平均欧氏速度误差为 e_v 时，速度得分为

$$S_{\mathrm{vel}} = \frac{1}{1+e_v}. \tag{8}$$

该映射使更大误差平滑趋零，保留 $[0,1]$ 得分范围。

E.3 Deduction Score Fusion

The final Deduction score combines MLLM and objective metrics in three groups:

$$S_{\text{integrity}} = \frac{1}{2}\left(S_{\text{ann_removed}} + S_{\text{object_consistency}}\right), \tag{9}$$

$$S_{\text{fidelity}} = \frac{1}{4}\left(S_{\text{visual_quality}} + S_{\text{motion_smoothness}} + S_{\text{PSNR}} + S_{\text{maskedPSNR}}\right), \tag{10}$$

$$S_{\text{physics}} = \frac{1}{5}\left(S_{\text{physics_accuracy}} + S_{\text{spatialIoU}} + S_{\text{stIoU}} + S_{\text{wIoU}} + S_{\text{vel}}\right). \tag{11}$$

The Deduction score is

$$S_{\text{Ded}} = 0.20S_{\text{integrity}} + 0.20S_{\text{fidelity}} + 0.60S_{\text{physics}}. \tag{12}$$

The larger physics weight reflects the benchmark objective: visual plausibility should not compensate for physically incorrect dynamics.

E.4 Hierarchical Aggregation

Scores are aggregated from rollout to case, from case to slice, and from slice to model. The three rollouts are first averaged as in Section D.6. For a model m and subtrack k, the track-wise score is

$$S_{m,k} = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} S_{m,c,k}. \tag{13}$$

The overall model score averages over cases and subtracks:

$$S_{m} = \frac{1}{|\mathcal{C}||\mathcal{K}|}\sum_{c\in\mathcal{C}}\sum_{k\in\mathcal{K}} S_{m,c,k}. \tag{14}$$

Pillar-wise and source-wise scores are computed by restricting $\mathcal{C}$ to the corresponding case subset. All reported cells are case-level means over the relevant subset, not means of previously reported table cells.

F Judge Reliability and Robustness

F.1 Primary Judge

The canonical MLLM judge is Gemini 3 Flash with temperature 0. The judge is used for response validity, format compliance, text rendering, visual quality, object consistency, velocity-arrow interpretation, and high-level physics plausibility. Programmatic metrics are used wherever masks, pixels, or velocities can be compared directly to GT.

F.2 Open-Weights Cross-Check

To verify that the leaderboard is not an artifact of a single judge, we re-evaluate a balanced subset of outputs using Qwen3-VL-30B-A3B-Instruct-FP8 as an open-weights judge. The cross-check covers seven representative candidate generation models and all five subtracks. The goal is ranking robustness rather than exact equality of absolute scores, since different judges may calibrate the $[0, 1]$ scale differently.

Table 14 reports the agreement. The two judges agree strongly on per-model average scores and preserve the same quality-tier ordering.

The residual disagreement is concentrated in two cases. First, the open-weights judge is sometimes more lenient toward visually plausible but law-violating motion. Second, it can score rubric items more independently, whereas the primary judge more consistently propagates earlier errors such as a wrong formula option into downstream formula and substitution criteria. Because the tier-level ranking is preserved, we treat Qwen3-VL as a reproducible robustness check and keep Gemini 3 Flash as the canonical judge for the main leaderboard.

E.3 推演评分融合

最终推演分按三组合并 MLLM 和客观指标：

$$S_{\text{integrity}} = \frac{1}{2}\left(S_{\text{ann_removed}} + S_{\text{object_consistency}}\right), \tag{9}$$

$$S_{\text{fidelity}} = \frac{1}{4}\left(S_{\text{visual_quality}} + S_{\text{motion_smoothness}} + S_{\text{PSNR}} + S_{\text{maskedPSNR}}\right), \tag{10}$$

$$S_{\text{physics}} = \frac{1}{5}\left(S_{\text{physics_accuracy}} + S_{\text{spatialIoU}} + S_{\text{stIoU}} + S_{\text{wIoU}} + S_{\text{vel}}\right). \tag{11}$$

推演得分为

$$S_{\text{Ded}} = 0.20S_{\text{integrity}} + 0.20S_{\text{fidelity}} + 0.60S_{\text{physics}}. \tag{12}$$

物理权重更高反映基准目标：视觉合理性不应补偿错误动力学。

E.4 层级汇总

评分由生成汇总至案例，再至切片，最后至模型。先按 D.6 平均三次生成。模型 m、子赛道 k 的赛道分为

$$S_{m,k} = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} S_{m,c,k}. \tag{13}$$

模型总分对案例、子赛道平均：

$$S_{m} = \frac{1}{|\mathcal{C}||\mathcal{K}|}\sum_{c\in\mathcal{C}}\sum_{k\in\mathcal{K}} S_{m,c,k}. \tag{14}$$

按支柱、来源的分数通过将 $\mathcal{C}$ 限制至对应案例子集计算。各单元均为相关子集的案例均值，非已报告表格单元的均值。

F 评审可靠性与稳健性

F.1 主评审模型

规范 MLLM 评审为温度 0 的 Gemini 3 Flash，用于响应有效性、格式、文本渲染、视觉质量、物体一致性、速度箭头解读及高层物理合理性。凡掩码、像素、速度可直接对真值比较，均使用程序指标。

F.2 开放权重交叉检查

为验证排行榜不只是单个评审的产物，用开放权重 Qwen3-VL-30B-A3B-Instruct-FP8 重新评估平衡输出子集，覆盖七个代表生成模型及全部五子赛道。目标是排名稳健性，而非绝对分精确相等，因为评审对 $[0, 1]$ 尺度可能校准不同。

表 14 报告一致性。两评审的逐模型均分高度一致，保留相同质量层级排序。

剩余分歧集中两种情况。第一，开放权重评审有时对视觉合理但违反定律的运动更宽松。第二，它可能更独立地评分量规各项，而主要评审更一致地将错误公式选项等早期错误传递至后续公式、代入标准。因层级排名保留，将 Qwen3-VL 作为可复现稳健性检查，Gemini 3 Flash 保持主要排行榜的规范评审。

Table 14: Agreement between Gemini 3 Flash and Qwen3-VL on per-model mean scores.

Comparison	Pearson r	Spearman ρ
Overall model average	0.95	0.93
Formulation-Text	0.99	1.00
Deduction	0.99	1.00
Formulation-Graphic	0.97	0.93
Perception-Text	0.81	0.96
Perception-Graphic	0.81	0.71

G Experimental Details

The benchmark evaluates 11 models: five video-generation models and six unified understanding-generation models. Each model is evaluated on 400 cases, five subtracks, and three independent rollouts, for 6000 responses per model. We do not tune prompts per model. Models receive the same annotated first frame and the same subtrack prompt template within their output family.

For the four non-Deduction subtracks, unified models directly generate the answer artifact and video models generate a chain-of-frames clip whose last frame is taken as the answer. For Deduction, video models generate a full clip and unified models generate timestamped keyframes at the evaluation timestamps.

H Limitations

Physics scope. Apple-π focuses on classical rigid-body mechanics: gravity-driven motion, momentum-conservation collisions, Newton's first law, and their simple compositions. It does not evaluate fluids, thermodynamics, electromagnetism, deformable bodies, fracture, granular media, biological motion, or quantum phenomena.

Object and scene scope. Dynamic objects are restricted to sphere, cube, cylinder, and cone. This improves annotation consistency but does not stress articulated, organic, deformable, or highly irregular objects. Cases use a single fixed camera, so the benchmark does not directly test multi-view consistency or novel-view generation.

Interface scope. The annotated first frame removes unnecessary reference ambiguity and makes physical quantities spatially grounded. As a result, Apple-π does not test whether a model can construct the full physical scene from text alone. It tests whether, given a visually grounded physical setup, the model can perceive quantities, formulate laws, and deduce dynamics.

Evaluation scope. MLLM judges are useful for response validity, format compliance, and perceptual qualities, but they are not perfect physical oracles. We therefore combine them with objective metrics whenever masks, pixels, or velocities can be compared to ground truth. Objective metrics also have limitations: mask IoU depends on segmentation quality, pixel metrics can penalize harmless appearance differences, and velocity estimation can be noisy for real-world clips.

Real-world ground-truth noise. Simulator cases have exact engine-state ground truth, but real-world cases rely on measurement, mask refinement, and frame-based annotation. We mitigate this with controlled recording, instrument precision, multi-pass review, and inter-annotator agreement analysis, but real-world labels remain less exact than simulator labels.

Temporal normalization assumption. For generated videos, we interpret the full decoded output as the requested physical duration in the prompt. This is necessary because providers may return fixed-length containers independent of the requested duration. The rule is consistent across all models, but it assumes that the generated clip represents the requested physical time rather than the provider's nominal file duration.

表 14：Gemini 3 Flash 与 Qwen3-VL 对逐模型平均分的一致性。

比较	Pearson r	Spearman ρ
模型整体平均分	0.95	0.93
定律表述—文本	0.99	1.00
推演	0.99	1.00
定律表述—图像	0.97	0.93
感知—文本	0.81	0.96
感知—图像	0.81	0.71

G 实验细节

基准评估五个视频生成、六个统一理解–生成模型，共 11 个。每模型 400 例、五子赛道、三次独立生成，共 6000 个响应。不为各模型调提示；同输出族内接收相同标注首帧、相同赛道模板。

四非推演赛道中，统一模型直接生成答案，视频模型生成帧链片段并取末帧。推演中视频模型生成完整片段，统一模型按评估时间戳生成关键帧。

H 局限性

物理范围。Apple-π 聚焦经典刚体力学：重力运动、动量守恒碰撞、牛顿第一定律及简单组合。不评流体、热力学、电磁学、可变形体、断裂、颗粒介质、生物运动、量子现象。

物体与场景范围。动态物体限定球、立方体、圆柱、圆锥，改善标注一致性，但不考验关节、有机、可变形或高度不规则物体。单固定相机案例不直接检验多视图一致性或新视角生成。

接口范围。标注首帧消除不必要指代歧义，使物理量有空间依据。因此 Apple-π 不检验仅从文本构建完整物理场景，而检验给定视觉化物理设置后能否感知量、表述定律、推演动力学。

评估范围。MLLM 可评响应有效性、格式、感知质量，但不是完美物理裁判。因此在掩码、像素、速度可与真值比较时，结合客观指标。客观指标也有限制：掩码 IoU 依赖分割质量，像素指标可能惩罚无害外观差异，真实片段速度估计可能有噪声。

真实真值噪声。仿真有精确引擎状态真值，真实案例依赖测量、掩码修正及逐帧标注。受控拍摄、仪器精度、多轮审核及标注员一致性分析减轻噪声，但真实标签仍不如仿真精确。

时间归一化假设。解码完整生成视频被解释为提示请求的物理时长，因为提供方可能返回与请求时长无关的固定长度容器。规则对全部模型一致，但假设片段代表请求物理时间，而非名义文件时长。